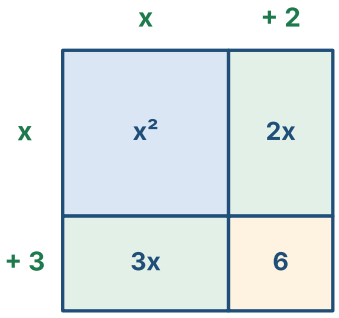
SAT Math 2.1 — Equivalent Expressions

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Multiplying and collecting

multiply every term by every term | then collect only like terms

Like terms share the same variable to the same power: $3x^2$ and $5x^2$ combine, $3x^2$ and $5x$ do not. The area model keeps every product in view and is worth drawing whenever a minus sign is involved.

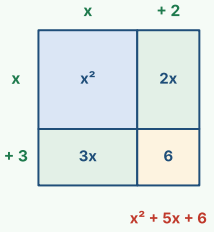


$(x + 2)(x + 3) = x^2 + 5x + 6$

The rectangle is (x + 2) wide and (x + 3) tall, so its area is the product. Each piece is one of the four multiplications, and the two green pieces are the like terms that collect into 5x.

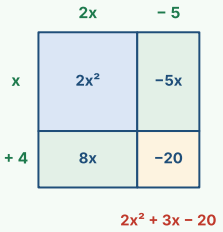
Example 1. **Expand $(x + 2)(x + 3)$.**

Solution.
$x \cdot x + 3x + 2x + 6$.
$= \mathbf{x^2 + 5x + 6}$.



Example 2. **Expand $(2x - 5)(x + 4)$.**

Solution.
$2x^2 + 8x - 5x - 20$.
$= \mathbf{2x^2 + 3x - 20}$.



Example 3. **Simplify $(3x^2 + 2x - 1) - (x^2 - 4x + 6)$.**

Solution.
Distribute the minus to *every* term: $3x^2 + 2x - 1 - x^2 + 4x - 6$.
$= \mathbf{2x^2 + 6x - 7}$.

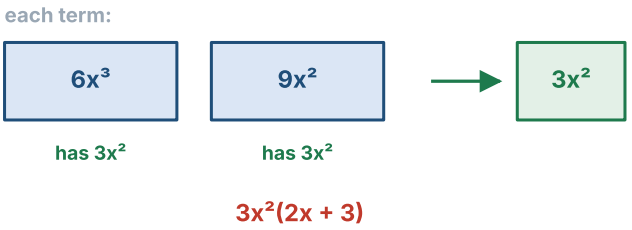
the minus belongs to every term in the bracket

$-(x^2 - 4x + 6)$	$= -x^2 + 4x - 6$
every sign inside flips	all three, not just the first

2 Common factor first, always

take out the greatest common factor before anything else

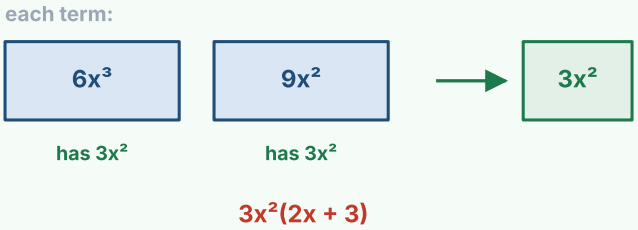
Pulling out the GCF turns an awkward expression into a familiar one and is the step most often skipped. Check the numbers and every variable: the GCF of $6x^3$ and $9x^2$ is $3x^2$, not 3.



Both terms carry three lots of x^2, so $3x^2$ comes out in front and what is left goes inside the bracket. Take the numbers *and* the letters — 3 alone would not be greatest.

Example 1. **Factor $6x^3 + 9x^2$ completely.**

Solution.
GCF is $3x^2$.
$= \mathbf{3x^2(2x + 3)}$.

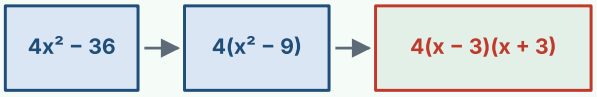


Example 2. **Factor $4x^2 - 36$ completely.**

Solution.

GCF 4 first: $4(x^2 - 9)$, and $x^2 - 9$ is a difference of squares.

= **$4(x - 3)(x + 3)$**. Stopping at $4(x^2 - 9)$ is not *completely* factored.



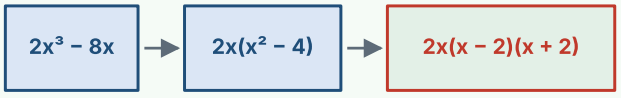
the GCF first, then the pattern underneath it

Example 3. **Factor $2x^3 - 8x$.**

Solution.

GCF $2x$: $2x(x^2 - 4)$.

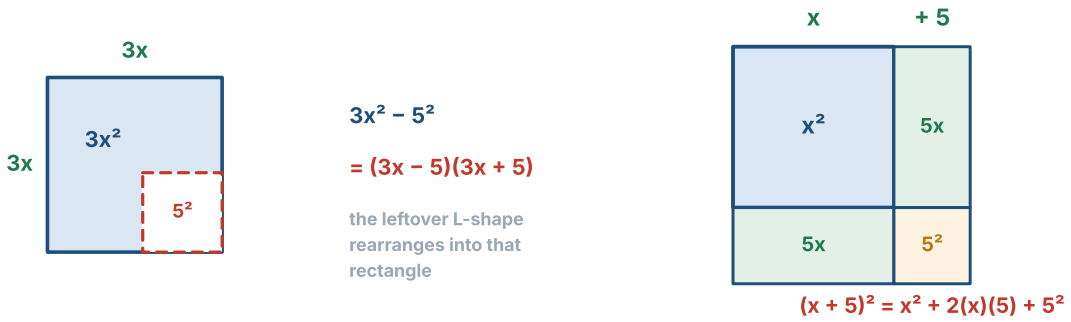
= **$2x(x - 2)(x + 2)$**.



3 Difference of squares and perfect squares

$$a^2 - b^2 = (a - b)(a + b) \mid a^2 \pm 2ab + b^2 = (a \pm b)^2$$

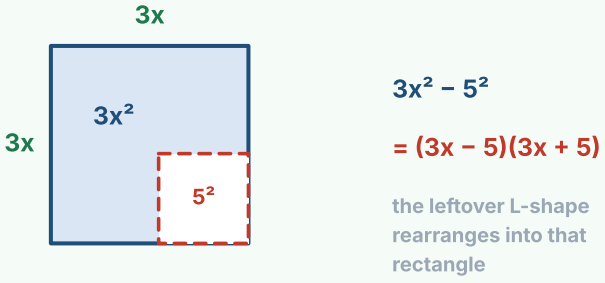
These two patterns appear constantly, often disguised. A sum of squares, $x^2 + 9$, does *not* factor over the real numbers — recognising that saves time on a question designed to waste it.



Left: cut a small square out of a big one and the L-shape left over rearranges into the rectangle (a − b)(a + b). Right: a perfect square has two identical middle pieces, which is where the 2ab comes from.

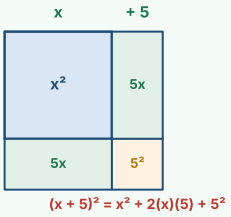
Example 1. **Factor $9x^2 - 25$.**

Solution.
This is $(3x)^2 - 5^2$.
$= \mathbf{(3x - 5)(3x + 5)}$.



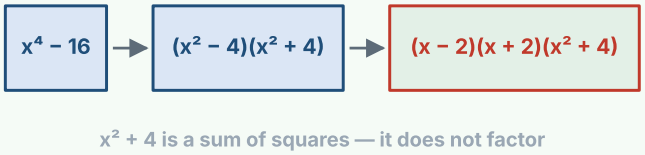
Example 2. **Factor $x^2 + 10x + 25$.**

Solution.
$25 = 5^2$ and $10x = 2(x)(5)$.
$= \mathbf{(x + 5)^2}$.



Example 3. **Factor $x^4 - 16$.**

Solution.
$(x^2)^2 - 4^2 = (x^2 - 4)(x^2 + 4)$, and the first bracket factors again.
$= \mathbf{(x - 2)(x + 2)(x^2 + 4)}$. The sum of squares stops here.

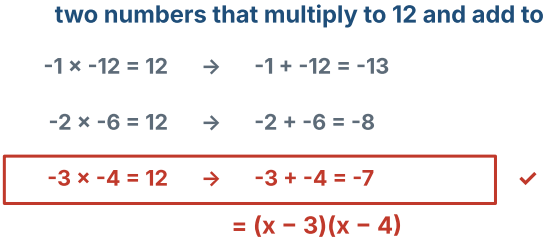


$x^2 + 4$ is a sum of squares — it does not factor

4 Factoring trinomials

$x^2 + bx + c$: find two numbers that multiply to c and add to b

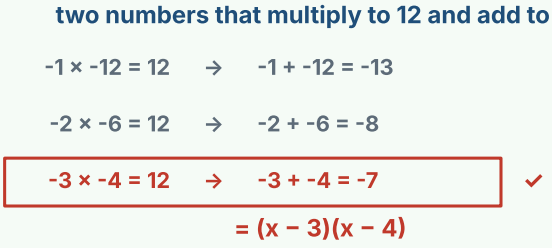
When the leading coefficient is not 1, multiply a by c, split the middle term using two numbers that multiply to ac and add to b, then factor by grouping. Writing out the factor pairs of ac beats guessing.



List the factor pairs and test the sum of each. Only one pair does both jobs, and that pair is the pair of brackets.

Example 1. **Factor $x^2 - 7x + 12$.**

Solution.
Two numbers multiplying to 12 and adding to −7: −3 and −4.
$= \mathbf{(x - 3)(x - 4)}$.



Example 2. **Factor $x^2 + 2x - 15$.**

Solution.

Multiply to −15, add to 2: 5 and −3.

= **(x + 5)(x − 3)**.

two numbers that multiply to −15 and add to 2

-1 × 15 = -15 → -1 + 15 = 14

1 × -15 = -15 → 1 + -15 = -14

3 × -5 = -15 → 3 + -5 = -2

5 × -3 = -15 → 5 + -3 = 2 ✓

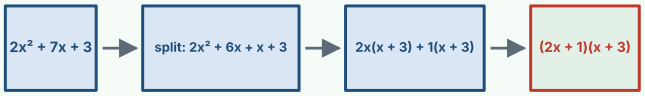
= (x + 5)(x − 3)

Example 3. **Factor $2x^2 + 7x + 3$.**

Solution.

ac = 6; two numbers multiplying to 6 and adding to 7 are 6 and 1.

$2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3) =$ **(2x + 1)(x + 3)**.



ac = 6, and 6 + 1 = 7

5 Exponent rules

$$x^m \cdot x^n = x^{m+n} \mid x^m / x^n = x^{m-n} \mid (x^m)^n = x^{mn} \mid x^{-n} = 1/x^n \mid x^0 = 1$$

Add exponents when multiplying, subtract when dividing, multiply when raising a power to a power. A negative exponent is a reciprocal, not a negative number: $2^{-3} = 1/8$, not -8.

five rules, and one worked instance of each

$x^m \cdot x^n = x^{m+n}$	$(3x^2y)(4x^3y^5) = 12x^5y^6$
$x^m / x^n = x^{m-n}$	$12^5/12^3 = 12^2 = 144$
$(x^m)^n = x^{mn}$	$(2x^3)^4 = 16x^{12}$
$x^{-n} = 1/x^n$	$5^{-2} = 1/25 = 0.04$
$x^0 = 1$	$7^0 = 1$

Read each rule with its example beside it. The fourth is the one that costs marks: a negative exponent flips the base, it never makes the value negative.

Example 1. **Simplify $(3x^2y)(4x^3y^5)$.**

Solution.
Multiply coefficients, add exponents.
= **$12x^5y^6$**.

coefficients: 3 × 4	= 12
x: 2 + 3	= x^5
y: 1 + 5	= y^6

Example 2. **Simplify $(2x^3)^4$.**

Solution.
Raise both parts to the fourth: $2^4 = 16$ and x^{12}.
= **$16x^{12}$**.

the 2 is inside the bracket too	$2^4 = 16$
exponents multiply	$(x^3)^4 = x^{12}$

Example 3. **Write 5^{-2} as a fraction.**

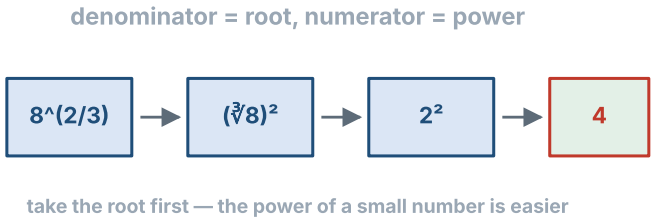
Solution.
1/25. A negative exponent flips the base; it never makes the value negative.

5^{-2}	$= 1/5^2 = 1/25 = 0.04$
not −25	the sign of the base never changes

6 Fractional exponents and radicals

$$x^{1/n} = \sqrt[n]{x} \mid x^{m/n} = \sqrt[n]{(x^m)}$$

The denominator of the exponent is the root; the numerator is the power. This conversion is the whole skill — once an expression is written with exponents, the ordinary exponent rules apply and radicals stop being special.



Denominator 3 means a cube root, numerator 2 means squared. Take the root first and you are squaring 2 instead of cube-rooting 64.

Example 1. **Write √x as a power of x.**

Solution.
$x^{1/2}$.

√x	= x^(1/2)
³√x	= x^(1/3)
ⁿ√x	= x^(1/n)

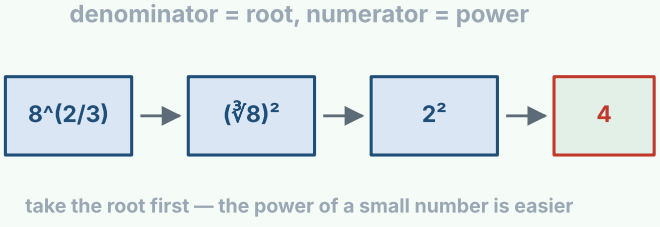
Example 2. **Simplify $x^{1/2} \cdot x^{1/3}$.**

Solution.
Add the exponents: 1/2 + 1/3 = 5/6.
= $x^{5/6}$.

same base → add the exponents	1/2 + 1/3 = 5/6
answer	x^(5/6)

Example 3. **Evaluate $8^{2/3}$.**

Solution.
Cube root first, then square: ³√8 = 2, and 2^2 = **4**. Taking the root first keeps the numbers small.



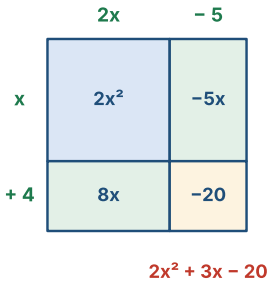
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

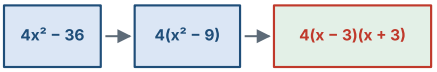
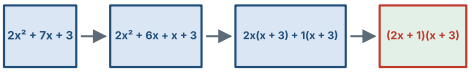
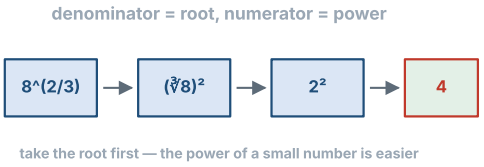
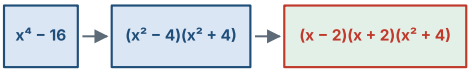
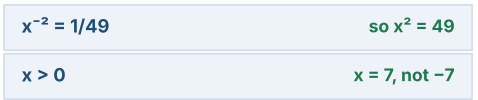
The SAT gives you Desmos on every maths question. For this topic it is **a checker, not a solver**. Desmos will not factor for you, but graphing your answer against the original shows instantly whether they are the same expression.

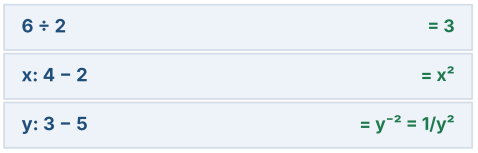
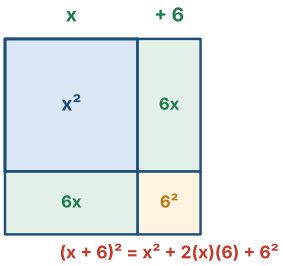
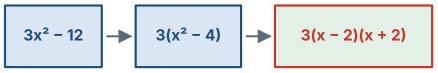
#	Question	Type into Desmos	Then	Answer
D1	Expand $(x + 2)(x + 3)$ and state the coefficient of x.	`(x+2)(x+3)`	Desmos will not expand symbolically; graph both forms to confirm they agree.	5
D2	What is the value of $8^{2/3}$?	`8^(2/3)`	Use the caret for exponents.	4
D3	What is the value of 5^{-2} as a decimal?	`5^(-2)`	A negative exponent gives a reciprocal.	0.04
D4	Simplify $12^5/12^3$.	`12^5/12^3`	Subtracting exponents gives 12^2.	144
D5	Evaluate $(2x^3)^4$ at x = 1.	`(2*1^3)^4`	Checks the coefficient $2^4 = 16$.	16
D6	At x = 5, what is the value of $x^2 - 7x + 12$?	`5^2-7*5+12`	Substitute to test a factorisation.	2
D7	At x = 3, does $(x - 3)(x - 4)$ equal zero?	`(3-3)*(3-4)`	A factor of zero makes the product zero.	0
D8	Evaluate $9x^2 - 25$ at x = 2.	`9*2^2-25`	Compare with $(3x - 5)(3x + 5)$ at x = 2.	11
D9	What is the value of $x^{1/2} \cdot x^{1/3}$ at x = 64?	`64^(1/2)*64^(1/3)`	Equals $64^{5/6}$.	32
D10	Evaluate $2x^3 - 8x$ at x = 2.	`2*2^3-8*2`	x = 2 is a zero, which confirms $(x - 2)$ is a factor.	0

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Read the word **completely** when it appears: more than one option may be equal in value, and only one is fully factored.

#	Level	Question	Answer	Solution
P1	Hard	Write an equivalent expression for $(2x - 5)(x + 4)$. 	$2x^2 + 3x - 20$	$8x - 5x = +3x$. $= 2x^2 + 3x - 20$. A common wrong answer multiplies only the first and last terms; C reverses the sign of the middle term.
P2	Hard	Write an equivalent expression for $(3x^2 + 2x - 1) - (x^2 - 4x + 6)$. the minus belongs to every term in the bracket $-(x^2 - 4x + 6)$ → $= -x^2 + 4x - 6$ every sign inside flips — all three, not just the first	$2x^2 + 6x - 7$	The minus sign applies to all three terms of the second bracket. $= 2x^2 + 6x - 7$. Two common wrong answers subtract only the first term.

#	Level	Question	Answer	Solution
P3	Hard	Factor completely: $4x^2 - 36$. 	**4(x − 3)(x + 3)**	Take the GCF 4 first, then the difference of squares. = **4(x − 3)(x + 3)**. Two common wrong answers equal in value but not *completely* factored — read the word 'complete' in the stem.
P4	Hard	What is the complete factorisation of $2x^2 + 7x + 3$? 	**(2x + 1)(x + 3)**	ac = 6, and 6 + 1 = 7, so split the middle term. 2x(x + 3) + 1(x + 3) = **(2x + 1)(x + 3)**. Expanding that answer gives $2x^2 + 7x + 3$.
P5	Hard	Write an equivalent expression for $(2x^3y^2)^3$. 	**$8x^9y^6$**	Every factor is cubed: $2^3 = 8$, and the exponents multiply by 3. = **$8x^9y^6$**. A common wrong answer multiplies the coefficient by 3 instead of cubing it.
P6	Hard	What is the value of $8^{2/3}$? 	**4**	The denominator is the root and the numerator is the power. $(\sqrt[3]{8})^2 = 2^2 =$ **4**.
P7	Very hard	Write an equivalent expression for $x^4 - 16$, factored completely over the real numbers. 	**(x − 2)(x + 2)(x² + 4)**	$x^4 - 16 = (x^2 - 4)(x^2 + 4)$, and $x^2 - 4$ factors again. = **(x − 2)(x + 2)(x² + 4)**. A sum of squares does not factor over the reals, so that answer is as far as it goes; A stops one step early.
P8	Very hard	If $x^{-2} = 1/49$ and $x > 0$, what is the value of x? 	**7**	$x^{-2} = 1/x^2$, so $x^2 = 49$. With $x > 0$, x = **7**.
P9	Very hard	Write an equivalent expression for $(x^{1/2})(x^{1/3})$ for $x > 0$. 	**$x^{5/6}$**	Multiplying powers of the same base adds the exponents. 1/2 + 1/3 = 5/6, so the answer is **$x^{5/6}$**. A common wrong answer multiplies the exponents.

#	Level	Question	Answer	Solution
P10	Very hard	The expression $(6x^4y^3)/(2x^2y^5)$ is equivalent to which of the following, for x and y nonzero? 	$3x^2/y^2$	Divide the coefficients and subtract the exponents: 6/2 = 3, $x^{4-2} = x^2$, $y^{3-5} = y^{-2}$. = **$3x^2/y^2$**. A common wrong answer leaves the negative exponent in the numerator as a positive one.
P11	Very hard	For what value of k is $x^2 + kx + 36$ a perfect square trinomial, given k > 0? 	12	A perfect square $(x + a)^2$ needs $a^2 = 36$, so a = 6, and the middle term is 2a. k = 2(6) = **12**.
P12	Very hard	The expression $3x^2 − 12$ can be written as $3(x − a)(x + a)$ for a positive constant a. What is the value of a? 	2	$3x^2 − 12 = 3(x^2 − 4) = 3(x − 2)(x + 2)$. So a = **2**. A common wrong answer takes a to be the 4 inside the bracket rather than its square root.

The one habit to build. Take the common factor out first, every single time. Half the ‘hard’ questions in this topic become one-liners once the GCF is gone.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify2.py; every figure is drawn to the numbers in the question.

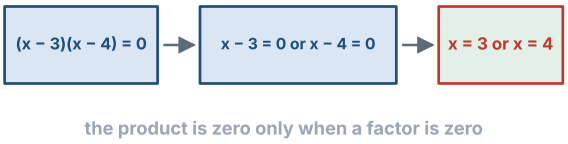
SAT Math 2.2 — Quadratic Equations

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

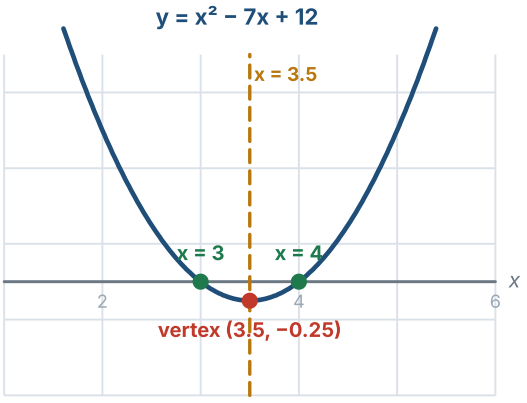
1 Solving by factoring

get = 0 first | factor | set each factor to zero

The zero-product property only works against zero, so move everything to one side before you factor. Dividing both sides by a variable loses a solution — factor it out instead.



the product is zero only when a factor is zero



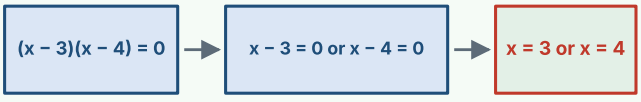
The factored form names the roots directly, and the graph shows them as the two places the curve meets the axis. Both brackets are zero nowhere else, which is why the method works.

Example 1. **Solve $x^2 - 7x + 12 = 0$.**

Solution.

$(x - 3)(x - 4) = 0$.

$x =$ **3 or 4**.



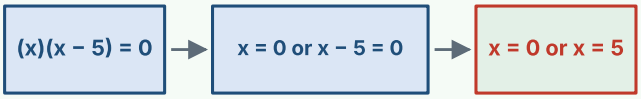
the product is zero only when a factor is zero

Example 2. **Solve $x^2 = 5x$.**

Solution.

$x^2 - 5x = 0$, so $x(x - 5) = 0$.

$x =$ **0 or 5**. Dividing by x at the start would have lost $x = 0$.



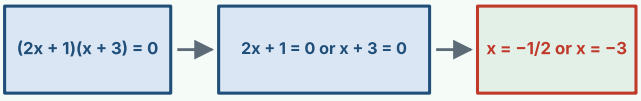
the product is zero only when a factor is zero

Example 3. **Solve $2x^2 + 7x + 3 = 0$.**

Solution.

$(2x + 1)(x + 3) = 0$.

$x =$ **$-1/2$ or -3**.



the product is zero only when a factor is zero

2 The quadratic formula

$x = [-b \pm \sqrt{(b^2 - 4ac)}] \div 2a$ — not on the reference sheet

Learn it: the SAT does not give it to you. Write a, b and c down before substituting, and keep the signs of b and c with their numbers. It works on every quadratic, factorable or not.

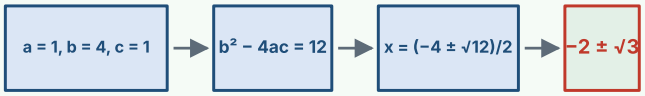
the formula works on every quadratic, factorable or not

$x = [-b \pm \sqrt{(b^2 - 4ac)}] \div 2a$	not on the reference sheet
write a, b and c down first	with their signs
$b^2 - 4ac$ is the discriminant	it decides how many roots

Three lines worth memorising. The sign of b travels with it: for $x^2 - 6x + 5$ the value of b is −6, so −b is +6.

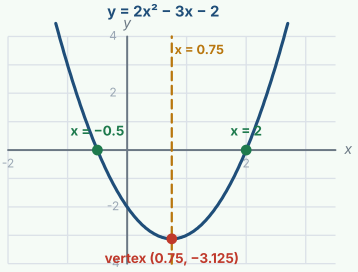
Example 1. **Solve $x^2 + 4x + 1 = 0$.**

Solution.
a = 1, b = 4, c = 1, so the discriminant is 16 − 4 = 12.
$x = (-4 \pm \sqrt{12})/2 =$ **$-2 \pm \sqrt{3}$**.



Example 2. **Solve $2x^2 - 3x - 2 = 0$.**

Solution.
Discriminant = 9 + 16 = 25, and √25 = 5.
x = (3 ± 5)/4, so x = **2 or −1/2**.



Example 3. **Why keep the sign with b?**

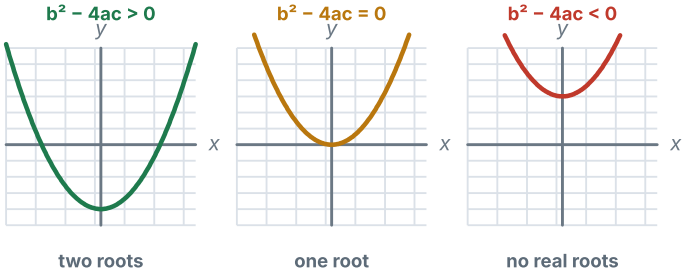
Solution.
For $x^2 - 6x + 5$, b = −6, so −b = +6. Writing b = 6 flips both roots — the single most common slip with the formula.

$x^2 - 6x + 5$	b = −6, so −b = +6
writing b = 6	gives −5 and −1 — both wrong

3 The discriminant

$b^2 - 4ac > 0$ → two real solutions | $= 0$ → one | < 0 → none

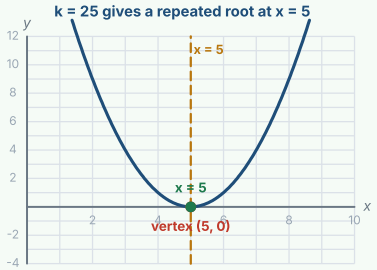
Whenever a question says ‘exactly one solution’, ‘a repeated root’, ‘touches the x-axis once’ or ‘the line is tangent’, it is asking you to set $b^2 - 4ac = 0$. One idea, many disguises.



The discriminant is the quantity under the root, so it counts the crossings: positive gives two, zero gives a touch, negative gives none because the root of a negative is not real.

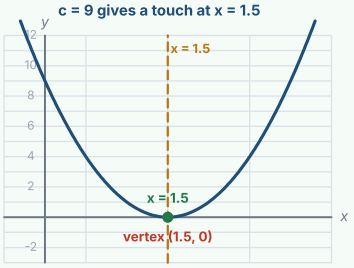
Example 1. **For what value of k does $x^2 - 10x + k = 0$ have exactly one solution?**

Solution.
$100 - 4k = 0$.
$k = \mathbf{25}$.



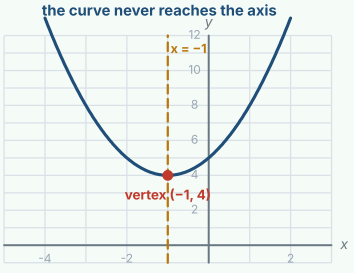
Example 2. **For what value of c does $4x^2 - 12x + c = 0$ have exactly one real solution?**

Solution.
$144 - 16c = 0$.
$c = \mathbf{9}$.



Example 3. **How many real solutions does $x^2 + 2x + 5 = 0$ have?**

Solution.
$4 - 20 = -16$, which is negative.
None.



4 Sum and product of the roots

sum = −b/a | product = c/a

When a question asks for the sum or product of the solutions, you never need the solutions themselves. Watch the sign: for $x^2 - 7x + 12$ the sum is +7, not −7.

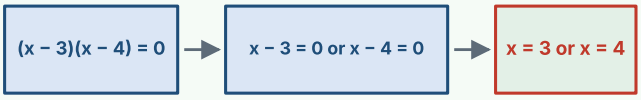
read them straight off the coefficients

sum of the roots	−b/a
product of the roots	c/a
$x^2 - 7x + 12$	sum −(−7)/1 = 7, product 12/1 = 12

The roots of $x^2 - 7x + 12$ are 3 and 4: they add to 7 and multiply to 12, exactly as −b/a and c/a predict — without solving anything.

Example 1. **What is the sum of the solutions of $x^2 - 7x + 12 = 0$?**

Solution.
−b/a = −(−7)/1 = **7**. Checking: 3 + 4 = 7.



the product is zero only when a factor is zero

Example 2. **What is the product of the solutions of $2x^2 + 7x + 3 = 0$?**

Solution.
c/a = 3/2 = **1.5**. Checking: (−1/2)(−3) = 1.5.

$2x^2 + 7x + 3$	a = 2, c = 3
product = c/a	3/2 = 1.5

Example 3. **The solutions of $x^2 + bx + 10 = 0$ sum to 7. What is b?**

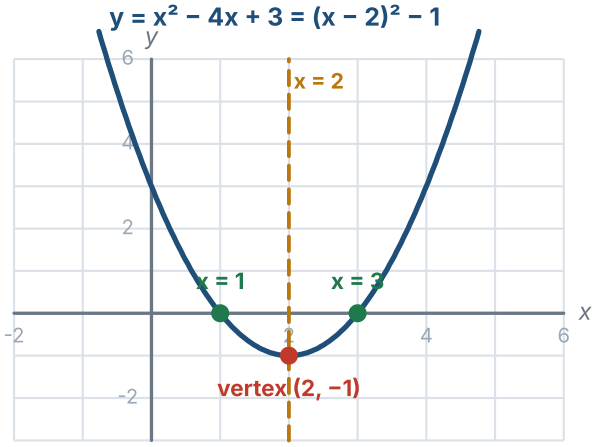
Solution.
−b = 7, so b = **−7**.

sum = −b/a = −b	and the sum is 7
so −b = 7	b = −7

5 Completing the square and vertex form

$x^2 + bx = (x + b/2)^2 - (b/2)^2$ | vertex form $y = a(x - h)^2 + k$ has vertex (h, k)

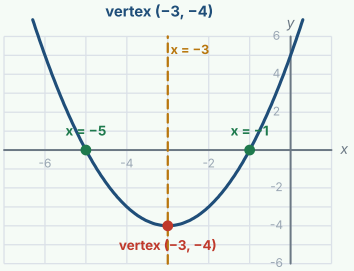
Halve the coefficient of x, square it, add and subtract it. Vertex form hands you the turning point directly, which is why the SAT asks you to produce it.



Both forms draw the same curve. Standard form shows the y-intercept 3; vertex form shows the turning point (2, −1) without any further work.

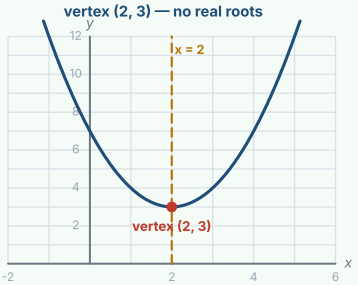
Example 1. **Write $x^2 + 6x + 5$ in vertex form.**

Solution.
Half of 6 is 3, and $3^2 = 9$.
$= (x + 3)^2 - 9 + 5 =$ **$(x + 3)^2 - 4$**, vertex $(-3, -4)$.



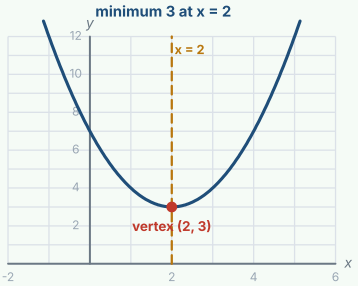
Example 2. **Write $x^2 - 4x + 7$ in vertex form.**

Solution.
Half of −4 is −2, squared is 4.
$=$ **$(x - 2)^2 + 3$**, vertex $(2, 3)$.



Example 3. **What is the minimum value of $y = x^2 - 4x + 7$?**

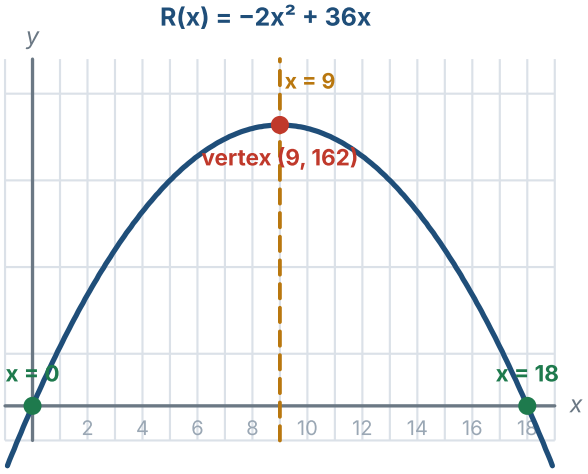
Solution.
From the vertex form, the smallest $(x - 2)^2$ can be is 0.
The minimum is **3**, reached at $x = 2$.



6 Quadratic word problems

maximum or minimum → the vertex | hits the ground / breaks even → a root

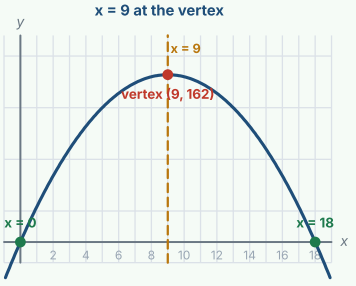
Translate the question into one of two requests. ‘What price gives the greatest revenue?’ wants the vertex’s x. ‘What is the greatest revenue?’ wants its y. Confusing those two costs marks every time.



The vertex (9, 162) answers both questions at once: x = 9 is the price that maximises revenue, y = 162 is the revenue itself. The roots are where revenue falls to zero.

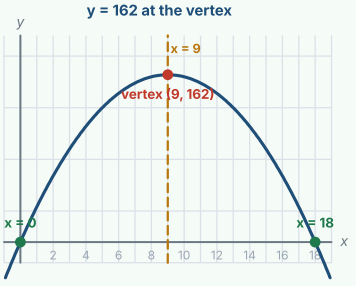
Example 1. **R(x) = −2x² + 36x. What price x gives the maximum revenue?**

Solution.
Vertex at x = −b/2a = −36/(−4) = **$9**.



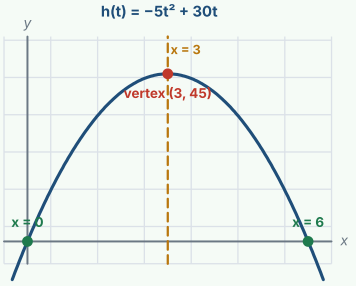
Example 2. **For that same R, what is the maximum revenue?**

Solution.
R(9) = −2(81) + 324 = **$162**. Note this answers a different question from the one above.



Example 3. **A ball’s height is h(t) = −5t² + 30t. When does it land?**

Solution.
−5t(t − 6) = 0, so t = 0 or 6.
It lands at **t = 6 seconds**; t = 0 is the launch.



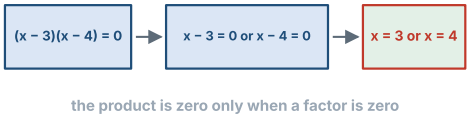
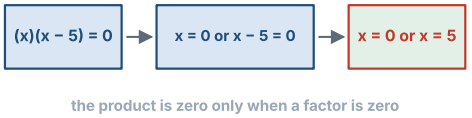
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

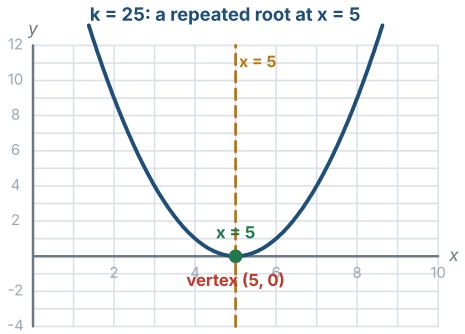
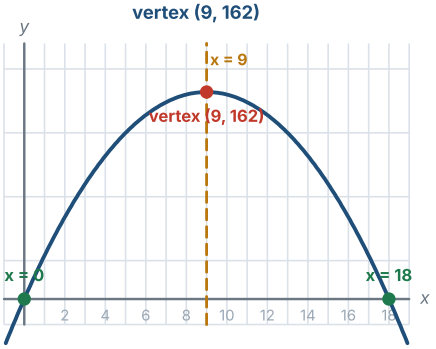
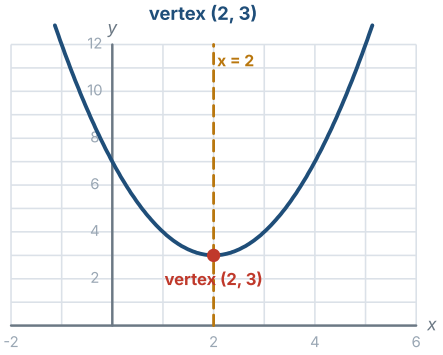
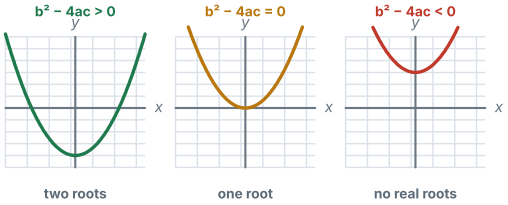
The SAT gives you Desmos on every maths question. Here it is **a root finder and a vertex finder**. Type the equation and read the x-intercepts, or type the function and click the turning point.

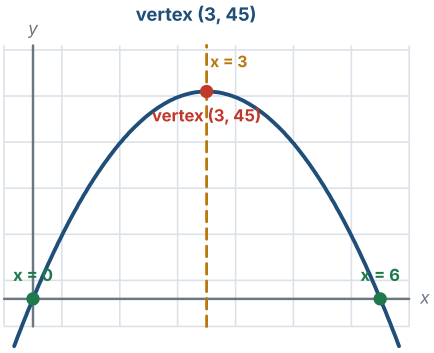
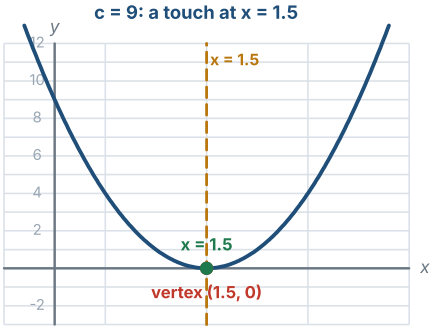
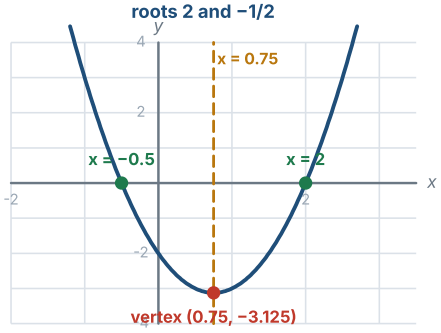
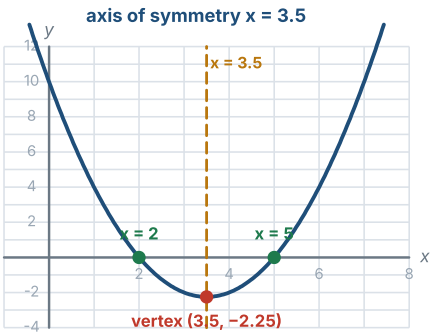
#	Question	Type into Desmos	Then	Answer
D1	What are the solutions of $x^2 - 7x + 12 = 0$?	`x^2-7x+12=0`	Type the equation; Desmos marks the roots on the x-axis.	3 and 4
D2	What is the sum of the solutions of $x^2 - 7x + 12 = 0$?	`3+4`	Or use −b/a directly.	7
D3	For what value of k does $x^2 - 10x + k = 0$ have exactly one solution?	`10^2/4`	Set the discriminant to zero: $k = b^2/4a$.	25
D4	What is the discriminant of $x^2 + 2x + 5 = 0$?	`2^2-4*1*5`	Negative means no real solutions.	-16
D5	Solve $2x^2 - 3x - 2 = 0$.	`2x^2-3x-2=0`	Graph it and read both x-intercepts.	2 and -0.5
D6	What is the minimum value of $y = x^2 - 4x + 7$?	`y=x^2-4x+7`	Graph it and click the turning point.	3
D7	$R(x) = -2x^2 + 36x$. What value of x gives the maximum?	`y=-2x^2+36x`	Click the vertex; Desmos labels it.	9
D8	For that R, what is the maximum value?	`-2*9^2+36*9`	The y of the vertex, not the x.	162
D9	At what positive time does $h(t) = -5t^2 + 30t$ return to zero?	`-5t^2+30t=0`	Read the right-hand root.	6
D10	What is the value of the discriminant of $4x^2 - 12x + 9$?	`12^2-4*4*9`	Zero means one repeated root.	0

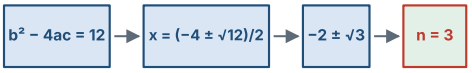
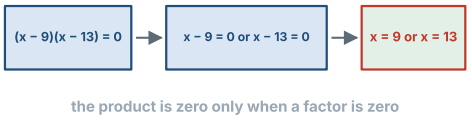
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Two questions decide most of these: is it asking **where** or **how much**, and does 'exactly one solution' appear anywhere in the stem?

#	Level	Question	Answer	Solution
P1	Hard	What is the sum of the solutions of the equation $x^2 - 7x + 12 = 0$? 	7	Sum of roots = −b/a = −(−7)/1 = **7**, and indeed 3 + 4 = 7. A common wrong answer applies the formula without the sign of b — the error five of your class made on Test 5.
P2	Hard	If $x^2 = 5x$ and $x \neq 0$, what is the value of x? 	5	$x^2 - 5x = 0$ gives $x(x - 5) = 0$, so x = 0 or 5. The condition $x \neq 0$ leaves **5**. Dividing by x at the start would also reach 5, but it silently discards a solution, which matters when the question does not rule it out.

#	Level	Question	Answer	Solution
P3	Hard	For what value of k does the equation $x^2 - 10x + k = 0$ have exactly one real solution? 	25	Exactly one solution means $b^2 - 4ac = 0$. $100 - 4k = 0$, so k = **25**. A common wrong answer is b/2 and B is b itself, both read off the equation rather than derived.
P4	Hard	A candle shop's daily revenue, in dollars, is $R(x) = -2x^2 + 36x$, where x is the price of one candle. What price gives the maximum daily revenue? 	$9	The vertex is at $x = -b/2a = -36/(-4) =$ **$9**. A common wrong answer is the maximum revenue R(9) = 162, which answers a different question; C is the coefficient read straight off the equation.
P5	Hard	Write an equivalent form of $x^2 - 4x + 7$. 	$(x - 2)^2 + 3$	Half of −4 is −2 and $(-2)^2 = 4$, so $x^2 - 4x = (x - 2)^2 - 4$. $= (x - 2)^2 + 3$. A common wrong answer forgets to subtract the 4 back out.
P6	Hard	How many distinct real solutions does $x^2 + 2x + 5 = 0$ have? 	Zero	$b^2 - 4ac = 4 - 20 = -16 < 0$. A negative discriminant means **no real solutions** — the parabola never reaches the x-axis.

#	Level	Question	Answer	Solution
P7	Very hard	A ball is thrown upward and its height, in metres, after t seconds is $h(t) = -5t^2 + 30t$. What is the maximum height reached, in metres? 	45	The vertex is at $t = -30/(-10) = 3$ seconds. $h(3) = -45 + 90 =$ **45 metres**. Answering 3 gives the time, not the height.
P8	Very hard	In the equation $4x^2 - 12x + c = 0$, c is a constant. If the equation has exactly one real solution, what is the value of c? 	9	$144 - 16c = 0$, so $c = 144/16 =$ **9**. A common wrong answer is 12/4, read off the equation rather than derived.
P9	Very hard	The solutions of $2x^2 - 3x - 2 = 0$ are p and q. What is the value of pq? 	−1	Product of roots $= c/a = -2/2 =$ **−1**. Checking: the roots are 2 and −1/2, and $2 \times (-1/2) = -1$. A common wrong answer uses c/a without the sign of c.
P10	Very hard	The graph of $y = x^2 + bx + 10$ has its vertex on the line x = 3.5. What is the value of b? 	−7	The vertex sits at $x = -b/2a = -b/2$. $-b/2 = 3.5$ gives $b =$ **−7**. A common wrong answer drops the minus sign.

#	Level	Question	Answer	Solution
P11	Very hard	The equation $x^2 + 4x + 1 = 0$ has solutions of the form $-2 \pm \sqrt{n}$, where n is a constant. What is the value of n? 	3	Discriminant = 16 − 4 = 12, so x = (−4 ± √12)/2 = −2 ± √12/2. √12/2 = 2√3/2 = √3, so n = **3**.
P12	Very hard	A rectangular garden has a perimeter of 44 metres and an area of 117 square metres. What is the length of its longer side, in metres? 	13	Half the perimeter is 22, so the sides are x and 22 − x with x(22 − x) = 117. $x^2 - 22x + 117 = 0$ factors as (x − 9)(x − 13) = 0, giving sides 9 and **13**. A common wrong answer is the shorter side.

The one habit to build. The quadratic formula is **not** on the reference sheet. Write a, b and c down with their signs before you substitute — losing the sign of b is the most expensive habit in this topic.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify2.py; every figure is drawn to the numbers in the question.

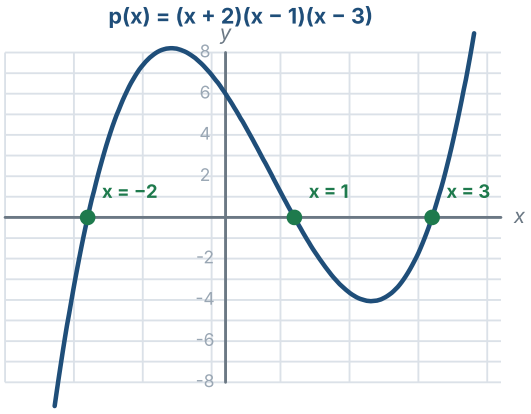
SAT Math 2.3 — Polynomials

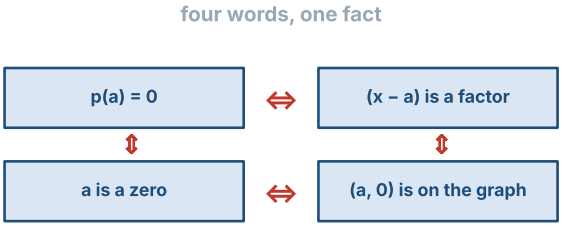
Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Zeros, roots, factors and intercepts

all four words mean the same thing: p(a) = 0 ⇔ (x − a) is a factor ⇔ (a, 0) is on the graph

The SAT swaps these words freely inside one question. Translate whichever you are given into whichever you need. A zero at x = 3 means the factor is (x − 3) — note the sign flip.

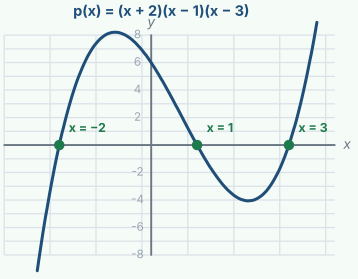




Each bracket switches off at one value of x, and that is where the curve meets the axis. The four statements on the right are the same fact in four vocabularies.

Example 1. **p(x) = (x + 2)(x − 1)(x − 3). What are its zeros?**

Solution.
Set each factor to zero.
x = **−2, 1 and 3**. Each sign flips.



Example 2. **A polynomial has zeros at x = −4 and x = 5. Give two of its factors.**

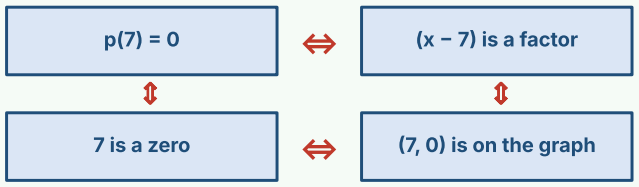
Solution.
(x + 4) and **(x − 5)**.

zero at x = −4	factor (x + 4)
zero at x = 5	factor (x − 5)
the sign always flips	x = a → (x − a)

Example 3. **The graph of p crosses the x-axis at (7, 0). What does that tell you about p?**

Solution.
p(7) = 0, and (x − 7) is a factor of p(x).

four words, one fact



2 The remainder and factor theorems

remainder on dividing by (x − a) is p(a) | remainder 0 ⇔ (x − a) is a factor

You never have to do the long division. To find the remainder, substitute. Watch the sign: dividing by (x + 2) means a = −2, because x + 2 = x − (−2).



and if p(a) = 0, the divisor is a factor

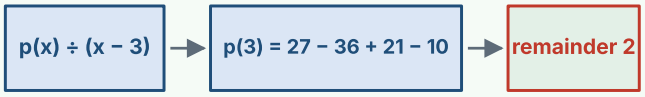
the sign flips on the way in

divide by (x − 3)	substitute x = +3
divide by (x + 2)	substitute x = −2

No division at all: put the number that switches the divisor off into p, and what comes out is the remainder. Zero means the divisor divides exactly.

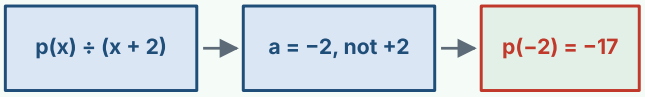
Example 1. **What is the remainder when $p(x) = x^3 - 4x^2 + 7x - 10$ is divided by x − 3?**

Solution.
p(3) = 27 − 36 + 21 − 10 = **2**.



Example 2. **What is the remainder when $p(x) = x^3 + 2x - 5$ is divided by x + 2?**

Solution.
Here a = −2.
p(−2) = −8 − 4 − 5 = **−17**.



Example 3. **Is $(x - 2)$ a factor of $x^3 - 3x^2 + 4$?**

Solution.

$p(2) = 8 - 12 + 4 = 0$.

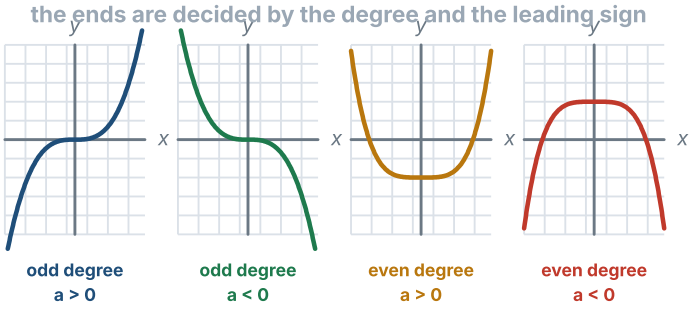
Yes — a remainder of zero means it is a factor.



3 Degree, end behaviour and turning points

degree n has at most n zeros and at most n − 1 turning points

The degree is the highest power. An odd-degree graph runs from bottom-left to top-right (or the reverse if the leading coefficient is negative); an even-degree graph has both ends pointing the same way.

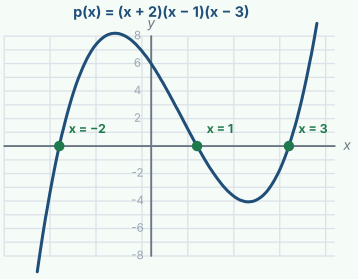


Four pictures cover every case. Odd degree sends the ends opposite ways; even degree sends them the same way; the leading sign says which way up.

Example 1. **How many turning points can a cubic have at most?**

Solution.

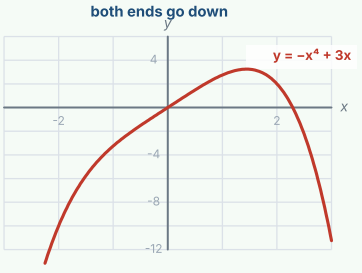
Degree 3, so at most **2** turning points.



Example 2. **Describe the end behaviour of $p(x) = -x^4 + 3x$.**

Solution.

Even degree with a negative leading coefficient, so **both ends go down**.



Example 3. **A polynomial has degree 5. What is the greatest number of x-intercepts it can have?**

Solution.

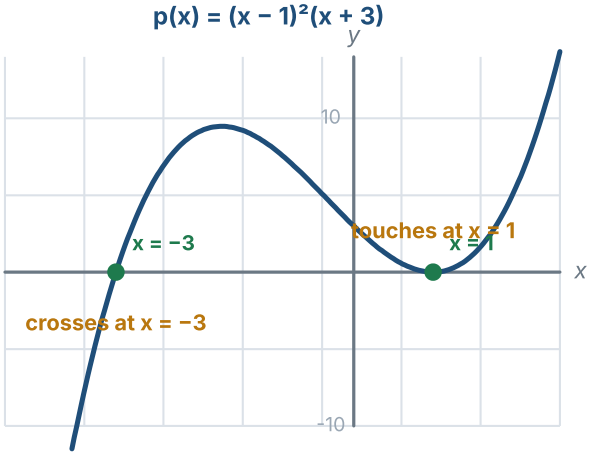
5. It could have fewer, but never more than its degree.

degree n	at most n zeros
degree n	at most n − 1 turning points
degree 5	5 zeros, 4 turning points at most

4 Multiplicity: crossing or touching

odd multiplicity → the graph crosses | even multiplicity → it touches and turns back

A repeated factor changes the shape at that zero. $(x - 2)^2$ makes the curve bounce off the axis at $x = 2$; $(x - 2)^3$ flattens but still passes through.

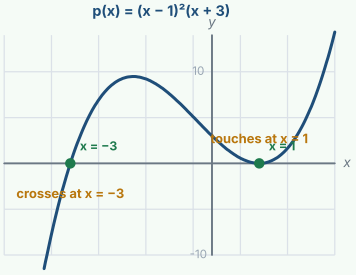


The squared factor at $x = 1$ turns the curve back without letting it through; the single factor at $x = -3$ lets it cross. Count the power on the bracket and you know which happens.

Example 1. **$p(x) = (x - 1)^2(x + 3)$. Describe the graph at each zero.**

Solution.

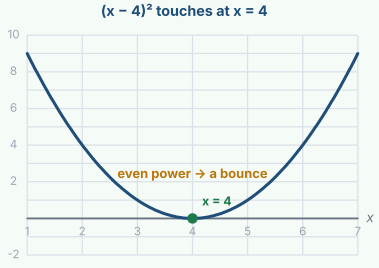
At $x = 1$ the multiplicity is 2, so it **touches and turns**. At $x = -3$ it is 1, so it **crosses**.



Example 2. **A graph touches the x-axis at $x = 4$ without crossing. What does that say about the factor?**

Solution.

$(x - 4)$ appears an **even** number of times — at least squared.



Example 3. **What is the degree of $p(x) = (x - 1)^2(x + 3)$?**

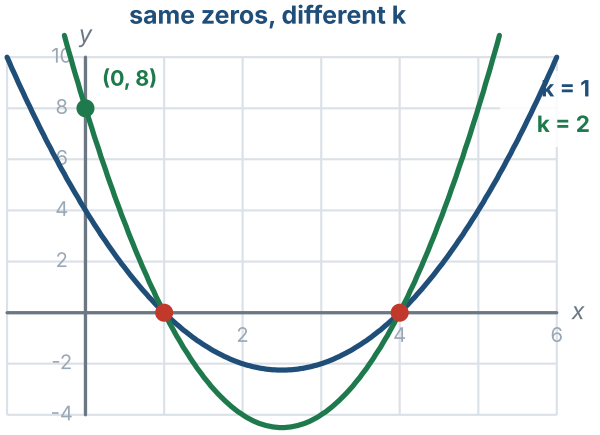
Solution.
Add the multiplicities: 2 + 1 = **3**.

$(x - 1)^2$	contributes 2
$(x + 3)$	contributes 1
degree	2 + 1 = 3

5 Building a polynomial from its zeros

zeros a, b, c → $p(x) = k(x - a)(x - b)(x - c)$

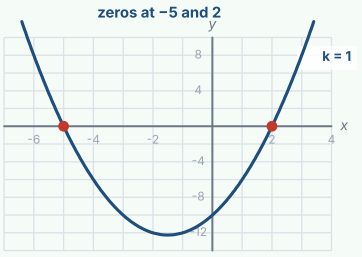
The zeros fix the shape but not the vertical scale, so a constant k remains. One extra point — usually the y-intercept — pins k down.



Both curves have zeros at 1 and 4; only the green one passes through (0, 8). That extra point is what fixes k = 2.

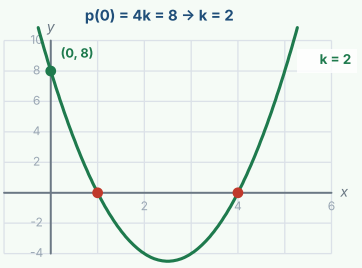
Example 1. **Write a quadratic with zeros 2 and −5.**

Solution.
$k(x - 2)(x + 5)$ for any nonzero k; with k = 1, $x^2 + 3x - 10$.



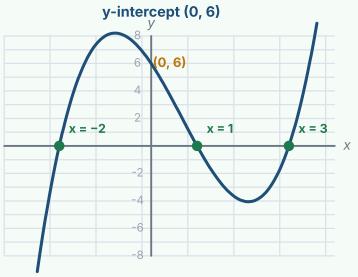
Example 2. **A quadratic has zeros 1 and 4 and passes through (0, 8). Find it.**

Solution.
$p(x) = k(x - 1)(x - 4)$, and $p(0) = 4k = 8$, so k = 2.
$p(x) = 2(x - 1)(x - 4)$.



Example 3. **What is the y-intercept of p(x) = (x + 2)(x − 1)(x − 3)?**

Solution.
p(0) = (2)(−1)(−3) = **6**.



6 Dividing polynomials

p(x) = (divisor)(quotient) + remainder

If a question gives you a quotient and a remainder, reassemble the original with this identity rather than dividing again. When the remainder is zero the divisor and quotient are factors.

reassemble instead of dividing again

p(x) = [x − 1] [x + 4] + [3]

= x² + 3x − 1

Multiply the divisor by the quotient and add the remainder: the original polynomial reappears, with no long division anywhere.

Example 1. **$x^2 + 5x + 6$ divided by (x + 2) gives what quotient?**

Solution.
$x^2 + 5x + 6 = (x + 2)(x + 3)$, so the quotient is **x + 3** with remainder 0.

reassemble instead of dividing again

p(x) = [x + 2] [x + 3] + [0]

= x² + 5x + 6

Example 2. **p(x) divided by (x − 1) gives quotient x + 4 and remainder 3. Find p(x).**

Solution.
p(x) = (x − 1)(x + 4) + 3 = $x^2 + 3x - 4 + 3$.
= **$x^2 + 3x - 1$**.

reassemble instead of dividing again

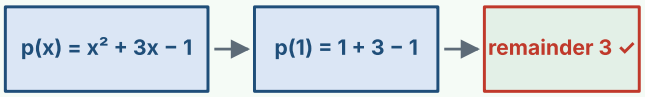
p(x) = [x − 1] [x + 4] + [3]

= x² + 3x − 1

Example 3. **Check that result using the remainder theorem.**

Solution.

p(1) = 1 + 3 − 1 = **3**, which is the stated remainder. The identity holds.



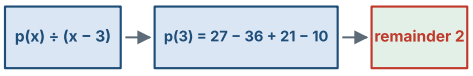
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

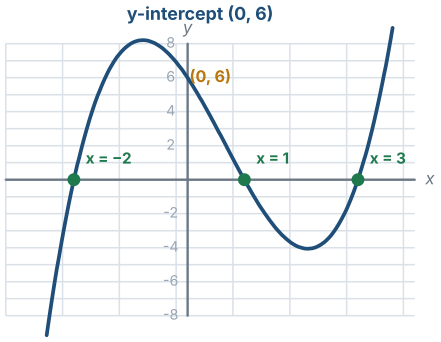
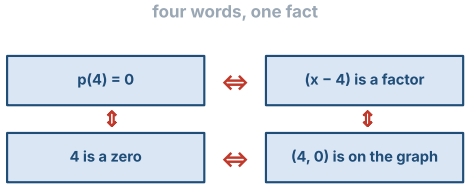
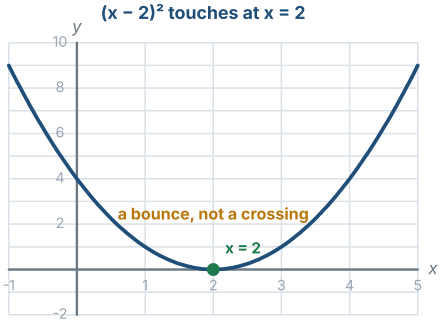
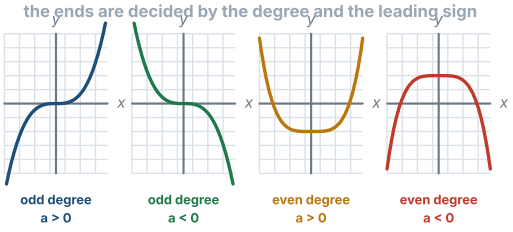
The SAT gives you Desmos on every maths question. For polynomials it is **a zero finder**. Graph p and read where it meets the axis; substitute a value to test a factor without dividing.

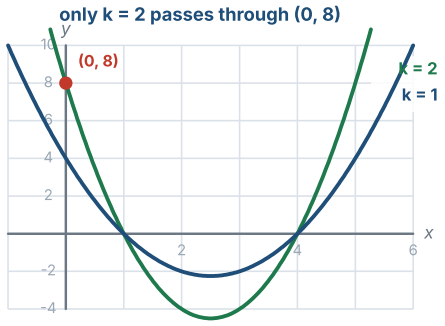
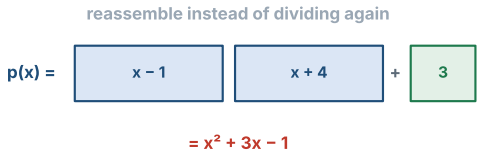
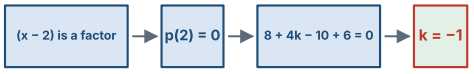
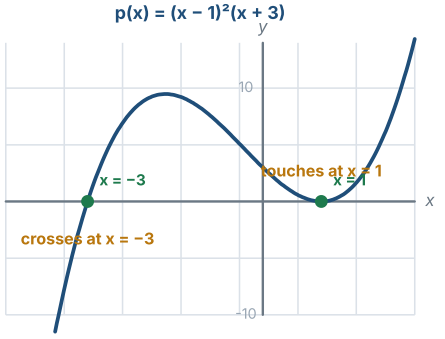
#	Question	Type into Desmos	Then	Answer
D1	What is the value of $x^3 - 4x^2 + 7x - 10$ at $x = 3$?	`3^3-4*3^2+7*3-10`	The remainder theorem in one line.	2
D2	What is the value of $x^3 + 2x - 5$ at $x = -2$?	`(-2)^3+2*(-2)-5`	Dividing by $x + 2$ means substituting -2.	-17
D3	Is $x = 2$ a zero of $x^3 - 3x^2 + 4$?	`2^3-3*2^2+4`	A value of 0 confirms the factor.	0 (yes)
D4	What is the y-intercept of $y = (x + 2)(x - 1)(x - 3)$?	`(0+2)*(0-1)*(0-3)`	Substitute $x = 0$.	6
D5	Graph $y = (x + 2)(x - 1)(x - 3)$. How many x-intercepts does it have?	`y=(x+2)(x-1)(x-3)`	Count where the curve meets the axis.	3
D6	For $p(x) = k(x - 1)(x - 4)$ with $p(0) = 8$, what is k?	`8/((0-1)*(0-4))`	Solve $4k = 8$.	2
D7	Expand $(x - 1)(x + 4) + 3$ at $x = 5$ to check a division.	`(5-1)*(5+4)+3`	Compare with $p(5)$ for $p(x) = x^2 + 3x - 1$.	39
D8	What is $x^2 + 3x - 1$ at $x = 5$?	`5^2+3*5-1`	Matches the line above, confirming the identity.	39
D9	Graph $y = (x - 1)^2(x + 3)$. At which zero does the curve only touch the axis?	`y=(x-1)^2(x+3)`	The squared factor makes it bounce.	x = 1
D10	What is the remainder when $x^4 - 1$ is divided by $x - 1$?	`1^4-1`	Zero means $(x - 1)$ is a factor.	0

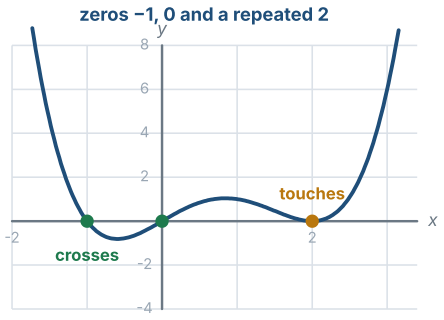
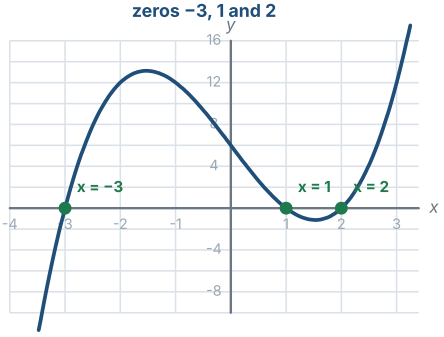
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Nearly every question here is the same idea in different words: **p(a) = 0** means $(x - a)$ is a factor means the graph crosses at $(a, 0)$.

#	Level	Question	Answer	Solution
P1	Hard	A shop's weekly profit index is modelled by $p(x) = x^3 - 4x^2 + 7x - 10$. What is the remainder when $p(x)$ is divided by $x - 3$? 	2	By the remainder theorem the answer is $p(3)$. $27 - 36 + 21 - 10 =$ **2**. No division is needed.

#	Level	Question	Answer	Solution
P2	Hard	The polynomial p is defined by $p(x) = (x + 2)(x - 1)(x - 3)$. What is the y-intercept of the graph of $y = p(x)$? 	6	The y-intercept is $p(0) = (2)(-1)(-3) = \mathbf{6}$. A common wrong answer loses one of the two negative signs.
P3	Hard	If $(x - 4)$ is a factor of the polynomial p, which of the following must be true? 	$p(4) = 0$	A factor $(x - a)$ corresponds to a zero at $x = a$, so $\mathbf{p(4) = 0}$. A common wrong answer flips the sign, which is the single most common error in this topic.
P4	Hard	The graph of a polynomial p touches the x-axis at $x = 2$ without crossing it. Which could be a factor of $p(x)$? 	$(x - 2)^2$	Touching without crossing means **even multiplicity**, so $(x - 2)^2$. Two common wrong answers use odd multiplicities, and both would cross.
P5	Hard	What is the greatest possible number of turning points on the graph of a polynomial of degree 5? 	4	A polynomial of degree n has at most $n - 1$ turning points. $5 - 1 = \mathbf{4}$.

#	Level	Question	Answer	Solution
P6	Hard	A quadratic function has zeros at x = 1 and x = 4 and passes through the point (0, 8). What is the value of the leading coefficient? 	2	p(x) = k(x − 1)(x − 4), so p(0) = k(−1)(−4) = 4k. 4k = 8 gives k = **2**.
P7	Very hard	When the polynomial p is divided by x − 1, the quotient is x + 4 and the remainder is 3. What is p(x)? 	$x^2 + 3x - 1$	p(x) = (x − 1)(x + 4) + 3 = x² + 3x − 4 + 3. = **x² + 3x − 1**. A common wrong answer forgets to add the remainder.
P8	Very hard	The polynomial p(x) = x³ + kx² − 5x + 6 has a factor of (x − 2). What is the value of k? 	−1	A factor (x − 2) means p(2) = 0. 8 + 4k − 10 + 6 = 0, so 4k + 4 = 0 and k = **−1**. Check: 8 − 4 − 10 + 6 = 0.
P9	Very hard	The graph of y = p(x) crosses the x-axis at x = −3, touches it at x = 1, and p has degree 3. Which could be p(x)? 	$(x + 3)(x - 1)^2$	Crossing at −3 needs (x + 3) to odd power; touching at 1 needs (x − 1) to even power; the degrees must total 3. = **(x + 3)(x − 1)²**. A common wrong answer swaps which zero is repeated; C has both signs wrong.
P10	Very hard	For p(x) = x³ + 2x − 5, what is the remainder when p(x) is divided by x + 2? 	−17	x + 2 = x − (−2), so substitute −2. −8 − 4 − 5 = **−17**. A common wrong answer substitutes +2 instead, which is the sign trap this question is built on.

#	Level	Question	Answer	Solution
P11	Very hard	A polynomial p of degree 4 has zeros at x = −1, 0, 2 and 2. State what is true here. 	**The graph crosses at x = −1 and x = 0 and touches at x = 2.**	x = 2 appears twice, so it has **even multiplicity** and the curve turns back there; the other two zeros are simple and the curve passes through. A common wrong answer confuses the degree with the number of distinct intercepts.
P12	Very hard	The polynomial p(x) = x^3 − 7x + 6 has x = 1 as a zero. What is the sum of its other two zeros? 	**−1**	Dividing by (x − 1) gives x^2 + x − 6, whose roots sum to −b/a = **−1** (they are 2 and −3). Alternatively all three zeros sum to 0 (no x^2 term), so the other two sum to 0 − 1 = −1.

The one habit to build. Watch the sign when you convert. A factor of (x + 2) means a zero at x = −2, and dividing by (x + 2) means substituting −2. Your class lost more marks to that flip than to any calculation.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify2.py; every figure is drawn to the numbers in the question.

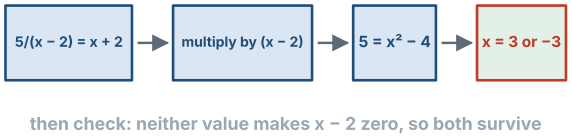
SAT Math 2.4 — Rational & Radical Equations

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1 Solving rational equations

multiply both sides by the denominator | then check nothing was made zero

Clearing the fraction turns the problem into something familiar. The one extra step is checking: any value that makes an original denominator zero must be thrown out, however correctly it was derived.



then check: neither value makes x − 2 zero, so both survive



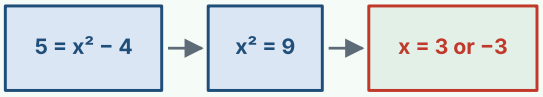
Clearing the denominator is a legal move, but it quietly forgets that x = 2 was never allowed. The number line keeps that exclusion in view while you solve.

Example 1. **Solve 5/(x − 2) = x + 2.**

Solution.

Multiply by (x − 2): 5 = (x + 2)(x − 2) = $x^2 - 4$.

$x^2 = 9$, so x = **3 or −3**. Neither makes x − 2 zero.



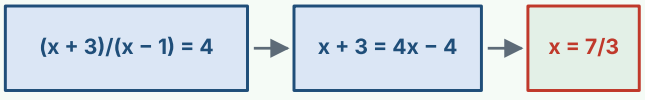
both check out

Example 2. **Solve (x + 3)/(x − 1) = 4.**

Solution.

x + 3 = 4(x − 1) = 4x − 4, so 7 = 3x.

x = **7/3**.



Example 3. **Solve 1/x + 1/(2x) = 3.**

Solution.

Multiply by 2x: 2 + 1 = 6x.

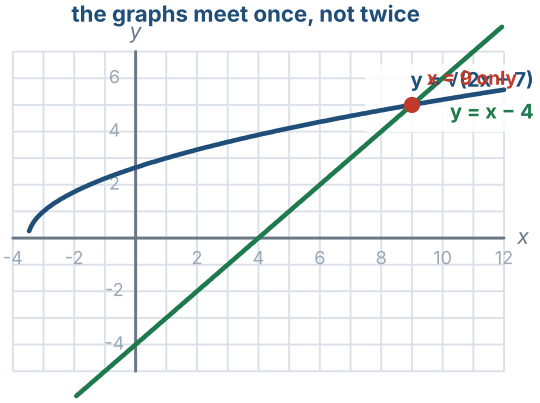
x = **1/2**. Check x ≠ 0, which it is not.



2 Extraneous solutions

squaring or clearing a denominator can invent solutions — always substitute back

This is the highest-value habit in the topic. Both squaring and multiplying out can produce values that satisfy the new equation but not the original. The check is part of the method, not an optional tidy-up.



substitute each candidate back into the original

x = 1	√9 = 3 against 1 − 4 = −3	× reject
x = 9	√25 = 5 against 9 − 4 = 5	✓ keep

Squaring turned one equation into another with an extra root. The graphs cross only at x = 9; x = 1 solves the squared equation and nothing else.

Example 1. **Solve √(2x + 7) = x − 4.**

Solution.

Squaring: 2x + 7 = x² − 8x + 16, so x² − 10x + 9 = 0 and x = 1 or 9.

Check x = 1: √9 = 3 but 1 − 4 = −3, so it fails. The solution set is **{9}**.

substitute each candidate back into the original

x = 1	√9 = 3 against 1 − 4 = −3	× reject
x = 9	√25 = 5 against 9 − 4 = 5	✓ keep

Example 2. **Why did x = 1 appear at all?**

Solution.
Squaring turns 3 = −3 into 9 = 9. The step is not reversible, so it can create solutions that were never there.

3 = −3	is false
square both sides	9 = 9 is true
so squaring	can invent a solution

Example 3. **Solve √(x + 5) = x − 1.**

Solution.
$x + 5 = x^2 - 2x + 1$ gives $x^2 - 3x - 4 = 0$, so x = 4 or −1.
x = −1 gives √4 = 2 but −2, so it fails. Answer **x = 4**.

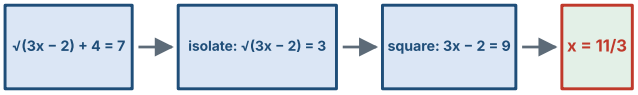
substitute each candidate back into the original

x = −1	√4 = 2 against −2	× reject
x = 4	√9 = 3 against 3	✓ keep

3 Radical equations

isolate the radical first, then square once

Get the root alone on one side before squaring, or you will square a sum and create cross terms. If two radicals appear, isolate one, square, then isolate and square again.

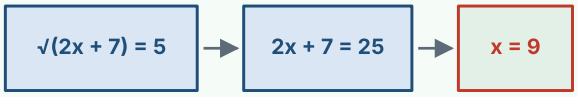


squaring before isolating leaves a radical behind

Four steps in order. Squaring the left side as it stands would give 3x − 2 + 8√(3x − 2) + 16 — still a radical, and worse than where you started.

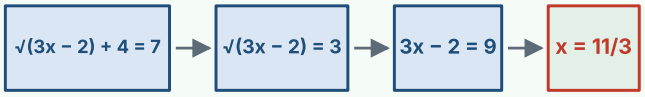
Example 1. **Solve √(2x + 7) = 5.**

Solution.
Square both sides: 2x + 7 = 25.
x = **9**. Check: √25 = 5.



Example 2. **Solve √(3x − 2) + 4 = 7.**

Solution.
Isolate first: √(3x − 2) = 3, so 3x − 2 = 9.
x = **11/3**.



Example 3. **Why not square √(3x − 2) + 4 = 7 immediately?**

Solution.
The left side would become 3x − 2 + 8√(3x − 2) + 16, which still has a radical in it — **worse than where you started**.

$(\sqrt{a} + 4)^2$	$= a + 8\sqrt{a} + 16$
the radical survives	so isolate it first

4 Domain restrictions

a denominator cannot be zero | an even root cannot take a negative

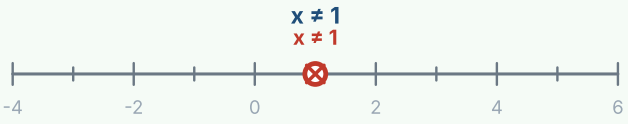
Questions often ask which values are excluded rather than asking you to solve anything. Set the denominator to zero and solve; for a square root, set the inside greater than or equal to zero.



A hollow crossed circle marks a value x may not take; a filled dot with a ray marks where a square root starts being defined. At x = 3 the root is √0 = 0, which is allowed.

Example 1. **For what value of x is (x + 3)/(x − 1) undefined?**

Solution.
x − 1 = 0.
x = 1.



Example 2. **What is the domain of $\sqrt{(2x - 6)}$?**

Solution.

$2x - 6 \geq 0$, so $2x \geq 6$.

$x \geq 3$.



Example 3. **For what values of x is $1/(x^2 - 9)$ undefined?**

Solution.

$x^2 - 9 = 0$ gives $x = \pm 3$.

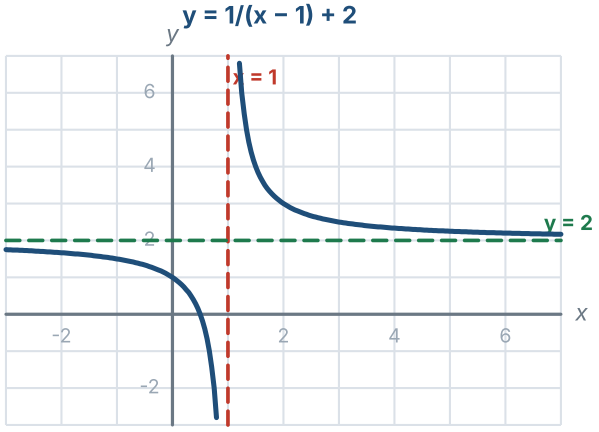
Undefined at **$x = 3$ and $x = -3$**.



5 Asymptotes of a rational function

vertical asymptote where the denominator is zero | horizontal asymptote from comparing degrees

A vertical asymptote is a value x cannot take. For the horizontal one: if the top's degree is smaller the asymptote is y = 0; if the degrees match it is the ratio of leading coefficients.

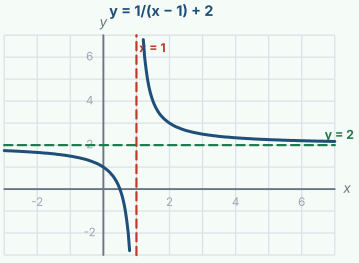


The curve runs away to infinity at x = 1, where the denominator vanishes, and flattens towards y = 2, the constant that was added on. Neither line is ever reached.

Example 1. **Find the vertical asymptote of $y = 1/(x - 1) + 2$.**

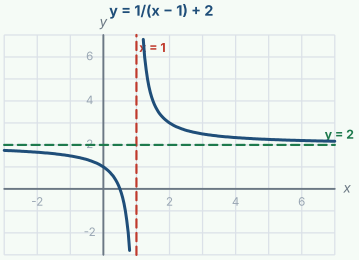
Solution.

The denominator vanishes at **$x = 1$**.



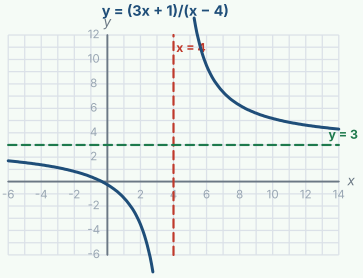
Example 2. **Find the horizontal asymptote of that same function.**

Solution.
As x grows, 1/(x − 1) tends to 0, so y tends to **2**.



Example 3. **Find the horizontal asymptote of y = (3x + 1)/(x − 4).**

Solution.
Degrees match, so take the ratio of leading coefficients.
y = 3.



6 Simplifying rational expressions

factor top and bottom, then cancel common factors — never cancel across a + sign

You may cancel a factor, never a term. $(x^2 + 3x)/x$ cancels to x + 3 because x is a factor of both terms on top; (x + 3)/x cancels to nothing at all.

cancel a factor, never a term

$$\frac{(x-3)\cancel{(x+3)}}{\cancel{(x+3)}} = x - 3$$

valid for $x \neq -3$

The bracket (x + 3) multiplies everything above and below it, so it cancels. A loose x added to something else does not multiply the whole line and cannot.

Example 1. **Simplify $(x^2 - 9)/(x + 3)$.**

Solution.
Factor the top: (x − 3)(x + 3)/(x + 3).
= **x − 3**, for x ≠ −3.

cancel a factor, never a term

$$\frac{(x-3)\cancel{(x+3)}}{\cancel{(x+3)}} = x - 3$$

$x \neq -3$

Example 2. **Simplify $(x^2 + 5x + 6)/(x^2 - 4)$.**

Solution.

$= (x + 2)(x + 3) / [(x - 2)(x + 2)].$

$=$ **$(x + 3)/(x - 2)$**, for $x \neq -2$.

cancel a factor, never a term

$$\frac{\cancel{(x+2)}\,(x+3)}{\cancel{(x+2)}} = (x+3)/(x-2), \quad x \neq -2$$

Example 3. **Can you cancel the x in (x + 4)/x?**

Solution.

No. On top x is a term, not a factor. The expression is already simplest.

$(x^2 + 3x)/x$	$= x + 3$ ✓ x is a factor
$(x + 4)/x$	cannot cancel — x is a term

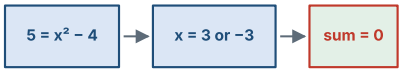
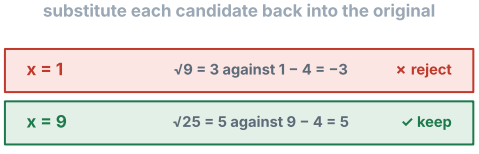
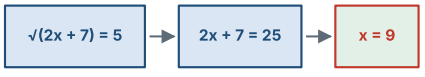
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

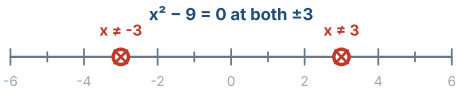
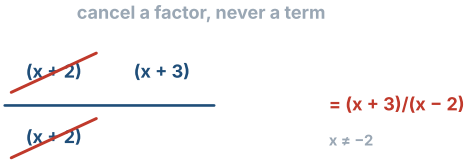
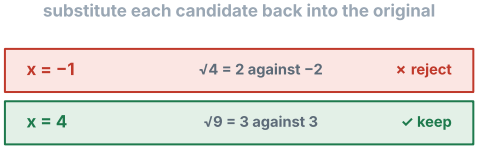
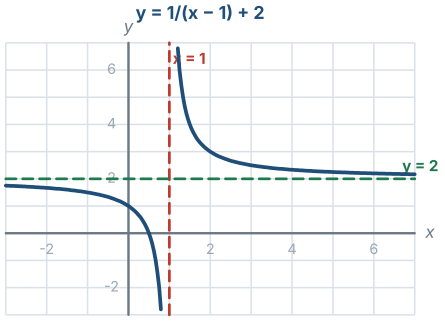
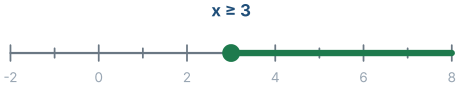
The SAT gives you Desmos on every maths question. Here it is **an extraneous-root filter**. Graph both sides: Desmos shows only the real intersections, so a root it does not show is one your algebra invented.

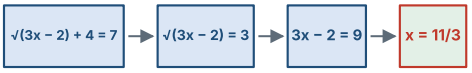
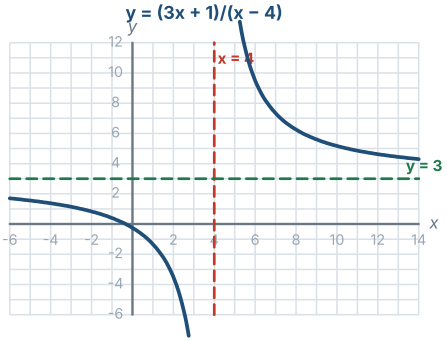
#	Question	Type into Desmos	Then	Answer
D1	Solve 5/(x − 2) = x + 2. What is the sum of the solutions?	`5/(x-2)=x+2`	Graph both sides; the roots are 3 and −3.	0
D2	Solve √(2x + 7) = x − 4.	`sqrt(2x+7)=x-4`	Desmos shows the genuine intersection, not the extraneous root.	9
D3	Solve √(2x + 7) = 5.	`sqrt(2x+7)=5`	Square both sides mentally to check.	9
D4	Solve (x + 3)/(x − 1) = 4.	`(x+3)/(x-1)=4`	Desmos gives the exact fraction.	7/3
D5	For what value of x is (x + 3)/(x − 1) undefined?	`x-1=0`	Graph it and look for the vertical asymptote.	1
D6	What is the horizontal asymptote of y = (3x + 1)/(x − 4)?	`y=(3x+1)/(x-4)`	Zoom out until the curve flattens.	3
D7	Solve 1/x + 1/(2x) = 3.	`1/x+1/(2x)=3`	Clear the denominators first if you prefer.	0.5
D8	Solve √(3x − 2) + 4 = 7.	`sqrt(3x-2)+4=7`	Isolate the radical first.	11/3
D9	Evaluate (x² − 9)/(x + 3) at x = 10.	`(10^2-9)/(10+3)`	Compare with x − 3 at x = 10.	7
D10	Solve √(x + 5) = x − 1.	`sqrt(x+5)=x-1`	Only one intersection appears; −1 is extraneous.	4

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Every radical equation on this sheet needs a check, and every rational one needs you to ask which values the denominator forbids.

#	Level	Question	Answer	Solution
P1	Hard	What is the sum of all solutions to the equation 5/(x − 2) = x + 2? 	0	Multiplying by (x − 2) gives 5 = x² − 4, so x² = 9 and x = ±3. 3 + (−3) = **0**. A common wrong answer is the positive root alone — the word *sum* means there is more than one.
P2	Hard	What is the solution set of the equation √(2x + 7) = x − 4? 	{9}	Squaring gives x² − 10x + 9 = 0, so x = 1 or 9. But x = 1 gives √9 = 3 against 1 − 4 = −3, which is false. Only **x = 9** survives. A common wrong answer is what you get by never checking — six of your class chose it on Test 5.
P3	Hard	If √(2x + 7) = 5, what is the value of x? 	9	Square both sides: 2x + 7 = 25, so 2x = 18 and x = **9**.

#	Level	Question	Answer	Solution
P4	Hard	The function f is defined by f(x) = (x + 3)/(x − 1). If f(a) = 4, what is the value of a? 	7/3	a + 3 = 4(a − 1) = 4a − 4, so 7 = 3a. a = **7/3**.
P5	Hard	For what value of x is the expression (x + 5)/(x² − 9) undefined, given x > 0? 	3	The expression is undefined where the denominator is zero: x² = 9 gives x = ±3. With x > 0, x = **3**.
P6	Hard	Write an equivalent expression for (x² + 5x + 6)/(x² − 4) for x ≠ ±2. 	(x + 3)/(x − 2)	Factor both: (x + 2)(x + 3) over (x − 2)(x + 2), and cancel (x + 2). = **(x + 3)/(x − 2)**.
P7	Very hard	What is the solution to √(x + 5) = x − 1? 	4	Squaring gives x² − 3x − 4 = 0, so x = 4 or −1. Testing x = −1: √4 = 2 but −1 − 1 = −2, so it fails. Only **x = 4** works. A common wrong answer keeps the extraneous root.
P8	Very hard	The graph of y = 1/(x − 1) + 2 has a vertical asymptote and a horizontal asymptote. What are they? 	x = 1 and y = 2	The denominator vanishes at x = 1; as x grows, 1/(x − 1) → 0, leaving y → 2. **x = 1 and y = 2.** A common wrong answer gives the asymptotes of y = 1/(x − 1) before the shift.
P9	Very hard	State the complete domain of f(x) = √(2x − 6). 	x ≥ 3	An even root needs a non-negative inside: 2x − 6 ≥ 0. **x ≥ 3.** The equals sign belongs: at x = 3 the value is √0 = 0, which is defined.
P10	Very hard	If 1/x + 1/(2x) = 3 and x ≠ 0, what is the value of x? 	1/2	Multiply through by 2x: 2 + 1 = 6x, so 6x = 3. x = **1/2**.

#	Level	Question	Answer	Solution
P11	Very hard	A student solves √(3x − 2) + 4 = 7 by squaring both sides immediately, obtaining 3x − 2 + 16 = 49. What is wrong, and what is the correct value of x? 	**The radical must be isolated first; x = 11/3.**	Squaring a sum is not squaring each piece: $(\sqrt{a} + 4)^2 = a + 8\sqrt{a} + 16$, so the radical survives. Isolating first gives √(3x − 2) = 3, then 3x − 2 = 9 and x = **11/3**.
P12	Very hard	What is the horizontal asymptote of the graph of y = (3x + 1)/(x − 4)? 	**y = 3**	The numerator and denominator have the same degree, so the asymptote is the ratio of leading coefficients. 3/1 = **y = 3**. A common wrong answer is the rule for when the top's degree is smaller.

The one habit to build. Substitute your answer back into the **original** equation. Squaring and clearing denominators are not reversible steps, and six of your class lost M2 Q25 on Test 5 by skipping the check.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify2.py; every figure is drawn to the numbers in the question.

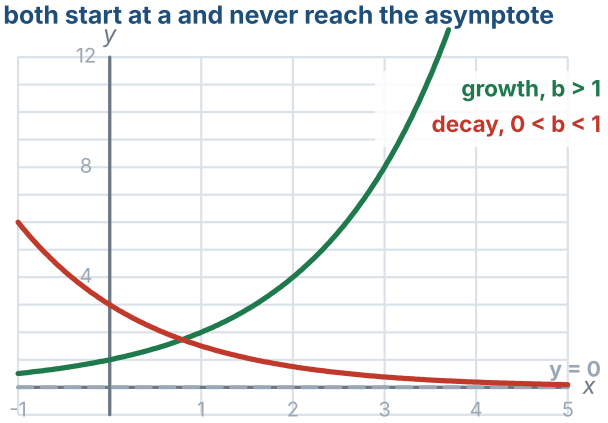
SAT Math 2.5 — Exponential Functions

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 The exponential model

$y = a \cdot b^x$ | a is the starting value, b is the growth factor per step

Read a at x = 0 and b as the multiplier for one step. Exponential means repeated multiplication, which is what separates it from a linear model's repeated addition.



Both curves cut the y-axis at their starting value a. Above b = 1 the curve climbs ever faster; below it the curve falls towards y = 0 without ever arriving.

Example 1. **A shop opens with 150 members and the number doubles each year. Write the model.**

Solution.
a = 150 and b = 2.
$M(t) = 150(2)^t$.



Example 2. **How many members after 3 years?**

Solution.
$150(2)^3 = 150 \times 8 =$ **1200**. Listing 150 → 300 → 600 → 1200 stops you counting one doubling short.

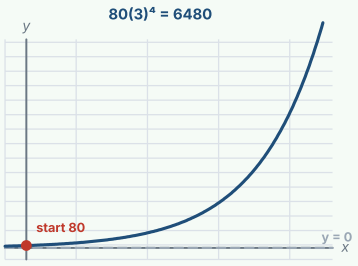
x = 0	x = 1	x = 2	x = 3
150	300	600	1200

differences	+150	+300	+600
ratios	×2	×2	×2

equal ratios → exponential

Example 3. **A culture starts at 80 and triples every hour. How many after 4 hours?**

Solution.
$80(3)^4 = 80 \times 81 = \mathbf{6480}$.



2 Growth, decay and the percentage

rise of r% → b = 1 + r | fall of r% → b = 1 − r

Convert between the base and the percentage in both directions. A base of 0.88 is a fall of 12%, not a fall of 88% and not 'keeps 12%'. A base of 1.07 is a rise of 7%.

the percentage is the distance from 1, never the base

rise of 15% →	b = 1 + 0.15 = 1.15
rise of 7% →	b = 1 + 0.07 = 1.07
fall of 8% →	b = 1 − 0.08 = 0.92
fall of 12% →	b = 1 − 0.12 = 0.88

The percentage is the distance from 1, never the base itself. 0.88 keeps 88% and therefore loses 12%.

Example 1. **A population falls 8% a year. What is the growth factor?**

Solution.
$1 - 0.08 = \mathbf{0.92}$.

the percentage is the distance from 1, never the base

fall of 8% → b = 1 − 0.08 = 0.92

Example 2. **V(t) = 4200(0.88)ᵗ. What does 0.88 mean?**

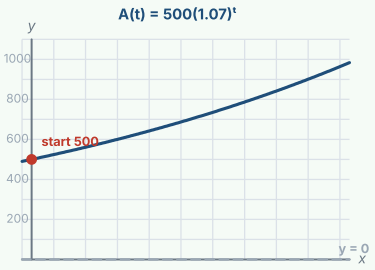
Solution.
The value **falls by 12% each year**, because 1 − 0.88 = 0.12. It keeps 88%, not 12%.

the percentage is the distance from 1, never the base

fall of 12% → b = 1 − 0.12 = 0.88

Example 3. **A deposit grows 7% a year. Write the model for $500.**

Solution.
$A(t) = 500(1.07)^t$.



3 Changing the time unit

$b^t = (b^{1/n})^{nt}$ — take the n-th root of the base, multiply the exponent by n

This is the subtopic your class scored lowest on. Changing from years to months never changes the base alone and never the exponent alone — it does both, in opposite directions. The tell in the question is a word like *monthly* or *quarterly*.

one year = 12 months

$V = a\cdot(0.88)^t$ → $a\cdot((0.88)^{(1/12)})^{12t}$

per year | per month

base → its 12-th root; exponent → ×12, both at once

$(0.88)^{(1/12)} \approx 0.9894$

Twelve monthly steps must come to one yearly step, so the monthly base is the twelfth root and the exponent is multiplied by twelve. Do one without the other and the model no longer matches.

Example 1. **$V(t) = 84{,}000(0.88)^t$ with t in years. Rewrite it with a monthly factor.**

Solution.
One year is 12 months, so t years is 12t months.
$84{,}000((0.88)^{1/12})^{12t}$, and $(0.88)^{1/12} \approx 0.9894$.

one year = 12 months

$V = a\cdot(0.88)^t$ → $a\cdot((0.88)^{(1/12)})^{12t}$

per year | per month

base → its 12-th root; exponent → ×12, both at once

$(0.88)^{(1/12)} \approx 0.9894$

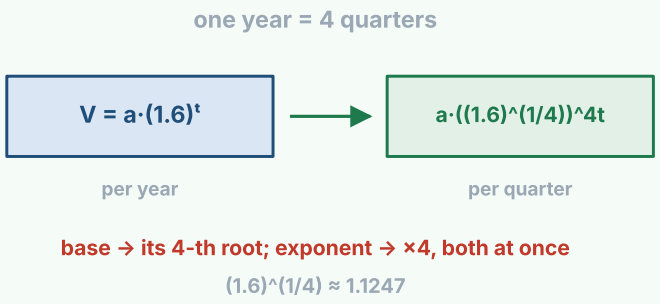
Example 2. **Why is $84{,}000(0.88)^{12t}$ wrong?**

Solution.
That multiplies by 0.88 twelve times a *year* — a fall of about 78% annually instead of 12%.

$(0.88)^{12}$	≈ 0.216 — a 78% fall
$(0.88^{1/12})^{12}$	$= 0.88$ ✓ the same 12% fall

Example 3. **P = 200(1.6)^t with t in years. Write it with a quarterly factor.**

Solution.
Four quarters in a year.
200((1.6)$^{1/4}$)^{4t}.



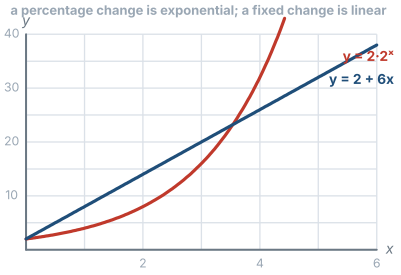
4 Exponential against linear

constant difference → linear | constant ratio → exponential

Test a table both ways. Subtract consecutive values: equal differences mean linear. Divide them: equal ratios mean exponential. Exponential growth eventually overtakes any linear growth, however slow it starts.



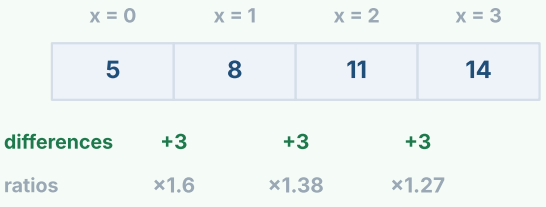




Same first value, two different engines. Adding a fixed amount draws a straight line; multiplying by a fixed amount bends upward and always wins in the end.

Example 1. **y = 5, 8, 11, 14. Which model?**

Solution.
Differences 3, 3, 3.
Linear, y = 3x + 2.



Example 2. **y = 5, 10, 20, 40. Which model?**

Solution.

Ratios 2, 2, 2.

Exponential, $y = 5(2)^x$ counting from x = 0.

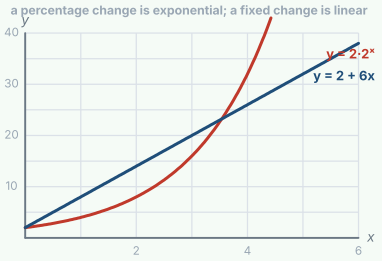
	x = 0	x = 1	x = 2	x = 3
	5	10	20	40
differences		+5	+10	+20
ratios		×2	×2	×2

equal ratios → exponential

Example 3. **A saver adds $50 a month; an investment grows 3% a month. Which is exponential?**

Solution.

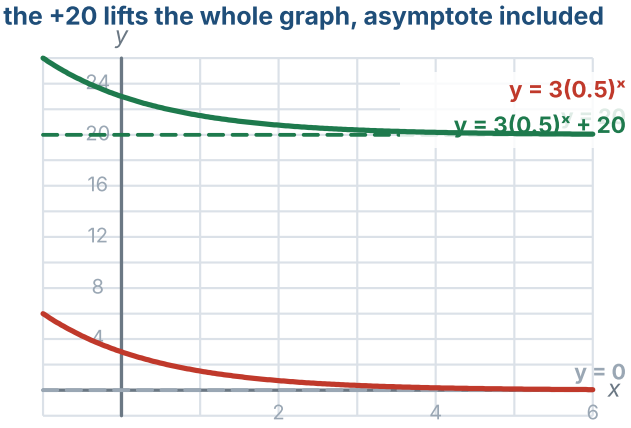
The **investment** — a percentage of a changing amount. Adding a fixed $50 is linear.



5 The horizontal asymptote

$y = a \cdot b^x$ approaches $y = 0$ | adding c shifts the asymptote to $y = c$

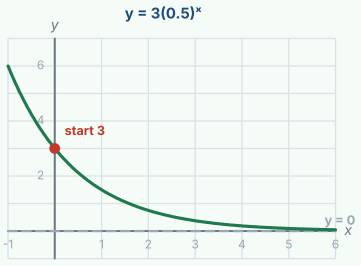
A decaying exponential gets arbitrarily close to its asymptote but never reaches it, which is why a model like 'cools towards room temperature' is written with a constant added on.



Both curves have the same shape; the green one sits 20 higher, so it levels off at 20 instead of 0. Whatever is added on is the value the model approaches.

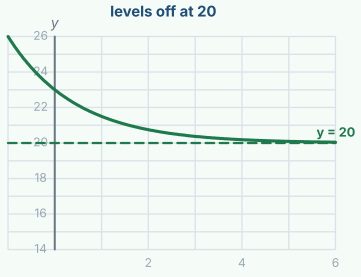
Example 1. **What is the horizontal asymptote of $y = 3(0.5)^x$?**

Solution.
$y = 0$. The curve falls towards the x-axis without touching it.



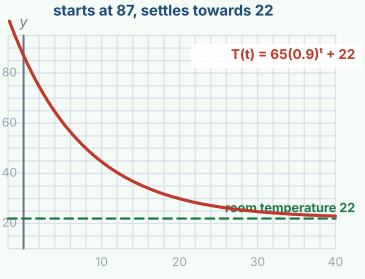
Example 2. **What is it for $y = 3(0.5)^x + 20$?**

Solution.
$y = 20$ — the whole graph is lifted 20.



Example 3. **A drink cools according to $T(t) = 65(0.9)^t + 22$. What does 22 represent?**

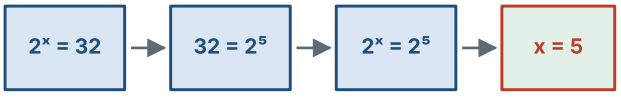
Solution.
The **room temperature**, 22 degrees — the value the drink approaches but never quite reaches.



6 Solving exponential equations

same base → equate the exponents | otherwise graph it in Desmos

Rewrite both sides as powers of the same number whenever you can. When the bases cannot be matched, the SAT expects you to read the intersection off a graph rather than use logarithms.

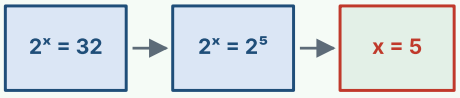


matching bases turns it into an ordinary equation

Once both sides are powers of 2, the exponents must agree. That is the whole method — no logarithms required.

Example 1. **Solve $2^x = 32$.**

Solution.
$32 = 2^5$.
$x = \mathbf{5}$.



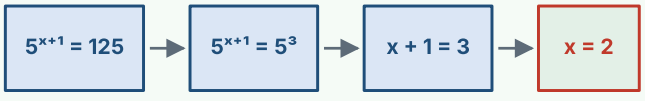
Example 2. **Solve $3^{2x} = 81$.**

Solution.
$81 = 3^4$, so $2x = 4$.
$x = \mathbf{2}$.



Example 3. **Solve $5^{x+1} = 125$.**

Solution.
$125 = 5^3$, so $x + 1 = 3$.
$x = \mathbf{2}$.



Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

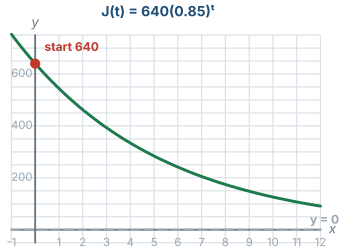
The SAT gives you Desmos on every maths question. Here it is **a growth calculator**. Type the whole model and evaluate it, or graph both sides of an exponential equation and read the crossing.

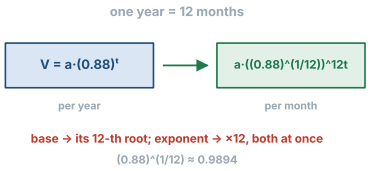
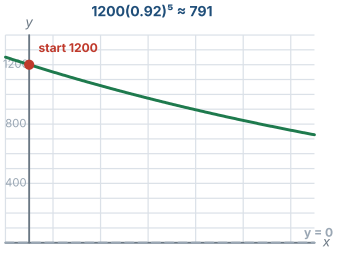
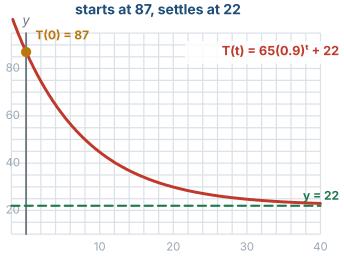
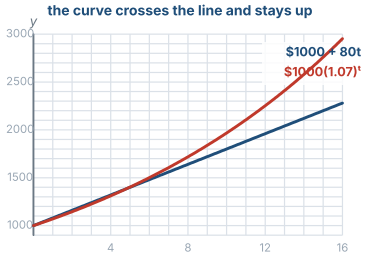
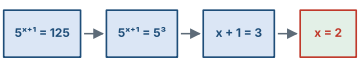
#	Question	Type into Desmos	Then	Answer
D1	A shop opens with 150 members and the number doubles each year. How many after 3 years?	`150*2^3`	Count the doublings carefully.	1200
D2	A town of 1200 falls 8% a year. What is the population after 5 years, to the nearest whole number?	`1200*0.92^5`	A fall of 8% means a factor of 0.92.	791
D3	What is $(0.88)^{1/12}$, to four decimal places?	`0.88^(1/12)`	The monthly factor behind an annual 0.88.	0.9894
D4	$500 grows at 7% a year. What is it worth after 3 years, to the nearest cent?	`500*1.07^3`	Repeated multiplication, not 500 × 1.21.	612.52
D5	Solve $2^x = 32$.	`2^x=32`	Graph both sides and read the intersection.	5
D6	A culture starts at 80 and triples hourly. How many after 4 hours?	`80*3^4`	Powers grow fast; let Desmos do it.	6480
D7	What is the growth factor for a rise of 15% per period?	`1+0.15`	A rise adds to 1.	1.15
D8	250 subscribers grow 20% monthly. How many after 2 months?	`250*1.2^2`	1.2 squared is 1.44.	360
D9	Solve $3^{2x} = 81$.	`3^(2x)=81`	Both sides are powers of 3.	2
D10	What does $65(0.9)^t + 22$ approach as t grows large?	`65*0.9^100+22`	Try a large t to see the asymptote.	22

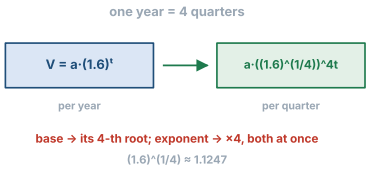
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Read the base carefully every time: it tells you growth or decay, and the percentage is its distance from 1 — not the base itself.

#	Level	Question	Answer	Solution
P1	Hard	When a tea shop opened it had 150 loyalty members, and the number has doubled every year since. How many members does the shop have 3 years after opening? 	1200	$150 \times 2^3 = 150 \times 8 =$ **1200**. A common wrong answer counts only two doublings — four of your class chose it on Test 5.

#	Level	Question	Answer	Solution
P2	Hard	The function $V(t) = 4200(0.88)^t$ gives the value, in talens, of an inventory t years after purchase. What is the best interpretation of 0.88? the percentage is the distance from 1, never the base fall of 12% → b = 1 − 0.12 = 0.88	**The value decreases by 12% each year.**	A base below 1 is decay, and 1 − 0.88 = 0.12. The value **falls 12% a year** and keeps 88%. A common wrong answer reverses which figure is kept.
P3	Hard	A spice stall opens with 640 jars and the number of unsold jars decreases by 15% each week. Which function gives the number of jars after t weeks? 	**$J(t) = 640(0.85)^t$**	A fall of 15% leaves 85%, so the base is 1 − 0.15 = 0.85. **$J(t) = 640(0.85)^t$.** A common wrong answer uses the rate as the base; C is linear.
P4	Hard	A subscription service has 250 subscribers and grows by 20% each month. How many subscribers will it have after 2 months? x = 0: 250 · x = 1: 300 · x = 2: 360 differences +50 +60 ratios ×1.2 ×1.2 equal ratios → exponential	**360**	$250(1.2)^2 = 250 \times 1.44 =$ **360**. Adding 20% of 250 twice would give 350, which is the common error.
P5	Hard	If $3^{2x} = 81$, what is the value of x? $3^{2x} = 81$ → $3^{2x} = 3^4$ → $2x = 4$ → $x = 2$	**2**	$81 = 3^4$, so $2x = 4$ and $x =$ **2**. A common wrong answer stops at $2x = 4$ without halving.
P6	Hard	A table shows y = 5, 10, 20, 40 for x = 0, 1, 2, 3. Which function models the data? x = 0: 5 · x = 1: 10 · x = 2: 20 · x = 3: 40 differences +5 +10 +20 ratios ×2 ×2 ×2 equal ratios → exponential	**$y = 5(2)^x$**	The ratios are all 2, so the model is exponential with a = 5 and b = 2. **$y = 5(2)^x$.** A common wrong answer swaps the starting value and the base; at x = 0 it gives 2, not 5.

#	Level	Question	Answer	Solution
P7	Very hard	The value V(t), in dollars, of a tour bus t years after purchase is V(t) = 84,000(0.88)^t. Which expression is equivalent and uses the approximate monthly decay factor? 	84,000$(0.88^{1/12})^{12t}$	One year is 12 months, so take the twelfth root of the base and multiply the exponent by 12. **84,000$(0.88^{1/12})^{12t}$**, where $0.88^{1/12} \approx 0.9894$. Two common wrong answers decay twelve times too fast; C treats 0.12 as a factor rather than a rate.
P8	Very hard	A population of 1,200 decreases by 8% each year. To the nearest whole number, what is the population after 5 years? 	791	$1200(0.92)^5 = 790.89\ldots \approx$ **791**. Subtracting 8% of 1,200 five times would give 720 — that treats the decrease as linear.
P9	Very hard	A hot drink cools according to T(t) = 65(0.9)^t + 22, where T is in degrees Celsius and t in minutes. What does the 22 represent? 	**The room temperature the drink approaches.**	As t grows, 65(0.9)^t → 0, so T → **22**, the horizontal asymptote. The drink's initial temperature is T(0) = 87, which is that answer's trap.
P10	Very hard	Two investments start at \$1,000. The first adds \$80 each year; the second grows by 7% each year. State what is true here. 	**The first leads at first, but the second eventually overtakes it.**	After one year: \$1,080 against \$1,070. But 7% of a growing balance eventually exceeds a fixed \$80. **Exponential growth always overtakes linear growth in the end**, whatever the head start.
P11	Very hard	If $5^{x+1} = 125$, what is the value of x? 	2	$125 = 5^3$, so x + 1 = 3. x = **2**.

#	Level	Question	Answer	Solution
P12	Very hard	A quantity P is modelled by $P = 200(1.6)^t$, where t is in years. Which expression gives P using a quarterly growth factor? 	$200(1.6^{1/4})^{4t}$	A year holds 4 quarters, so take the fourth root of the base and multiply the exponent by 4. **$200(1.6^{1/4})^{4t}$.** A common wrong answer is algebraically equal to the original but uses an *annual* factor raised to a quarter power — it is not the quarterly factor the question asks for.

The one habit to build. Changing the time unit changes **both** the base and the exponent, in opposite directions. Your class scored 23% on that one subtopic, the second worst on the whole paper, so drill it until the shape of the answer is automatic.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify2.py; every figure is drawn to the numbers in the question.

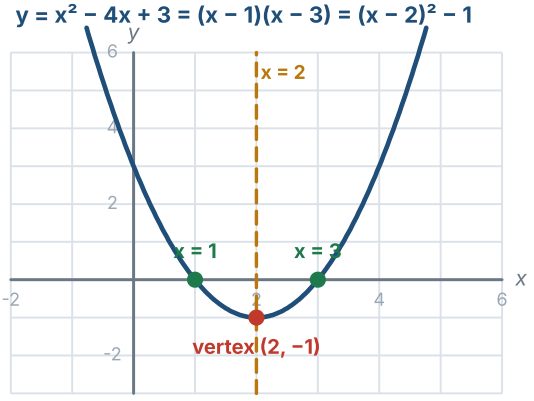
SAT Math 2.6 — Quadratic & Polynomial Graphs

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Three forms, three facts

standard ax^2+bx+c → y-intercept c | factored → x-intercepts | vertex $a(x-h)^2+k$ → vertex (h, k)

Each form hands you one fact for free. The question tells you which fact it wants, and that tells you which form to produce. Converting between them is the work; knowing which to aim for is the skill.



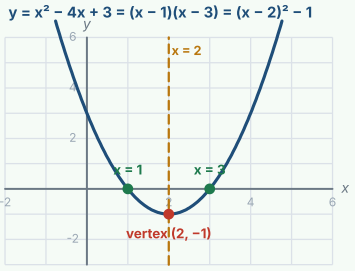
one curve, three ways of writing it

standard: $x^2 - 4x + 3$	y-intercept (0, 3)
factored: $(x - 1)(x - 3)$	x-intercepts 1 and 3
vertex: $(x - 2)^2 - 1$	vertex (2, −1)

All three expressions describe the identical parabola. Which one you write depends only on which number the question is asking for.

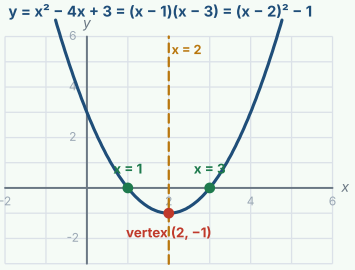
Example 1. **$y = x^2 - 4x + 3$. What is the y-intercept?**

Solution.
Standard form gives it immediately: c = **3**, so (0, 3).



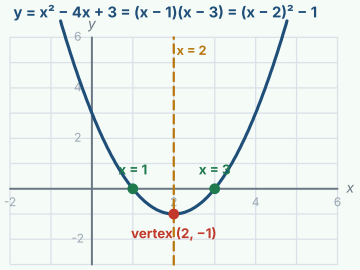
Example 2. **Write the same function in factored form and give the x-intercepts.**

Solution.
$(x - 1)(x - 3)$, so the x-intercepts are **(1, 0) and (3, 0)**.



Example 3. **Write it in vertex form and give the vertex.**

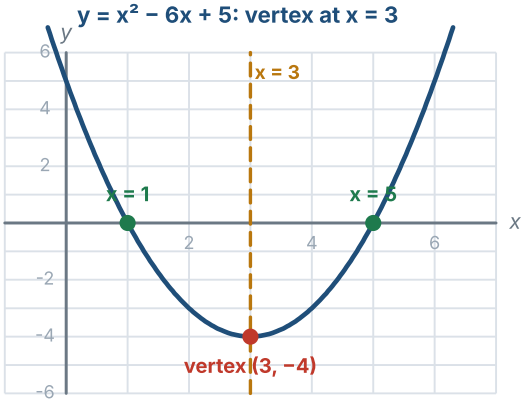
Solution.
$(x - 2)^2 - 1$, so the vertex is **(2, −1)**.



2 Finding the vertex

x = −b/2a | or the midpoint of the two x-intercepts

Both routes give the same x. Substitute it back to get the y. When a question gives you the factored form, the midpoint of the roots is faster than the formula.



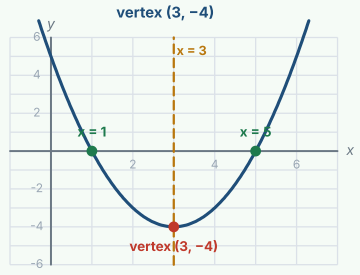
two routes, one answer

from standard form	x = −b/2a = 6/2 = 3
from the roots 1 and 5	midpoint (1 + 5)/2 = 3
then substitute	y = 9 − 18 + 5 = −4

The parabola is symmetric about its vertex, so the turning point sits exactly halfway between the two crossings — which is why both routes agree.

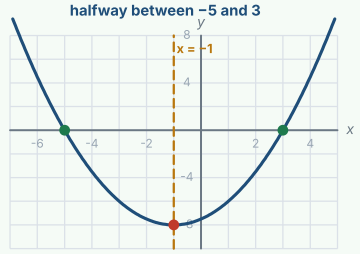
Example 1. **Find the vertex of $y = x^2 - 6x + 5$.**

Solution.
x = 6/2 = 3, and y = 9 − 18 + 5 = −4.
Vertex **(3, −4)**.



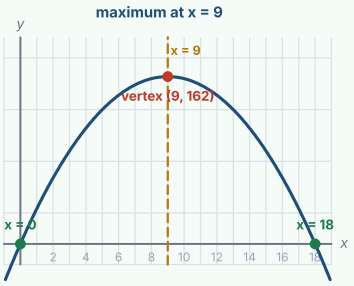
Example 2. **Find the vertex of $y = a(x - 3)(x + 5)$.**

Solution.
Midpoint of 3 and −5 is (3 − 5)/2 = **x = −1**.



Example 3. **R(x) = −2x² + 36x. Find the x that maximises R.**

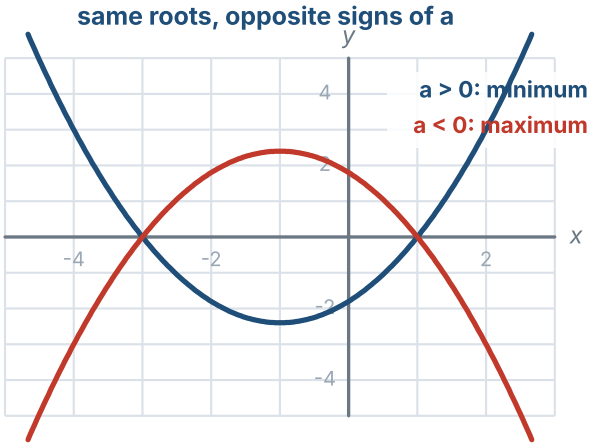
Solution.
x = −36/(2 × −2) = **9**.



3 Which way does it open, and what does that mean?

a > 0 → opens up, vertex is a minimum | a < 0 → opens down, vertex is a maximum

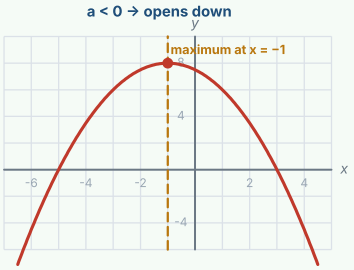
The sign of a settles maximum against minimum on its own. A negative a does not make everything else negative — the y-intercept, for instance, can still be positive.



Flipping the sign of a turns the curve over: the lowest point becomes the highest. The roots stay exactly where they were.

Example 1. **y = a(x − 3)(x + 5) with a < 0. Does the vertex give a maximum or a minimum?**

Solution.
a < 0 means the parabola opens downward, so the vertex is a **maximum**, at x = −1.



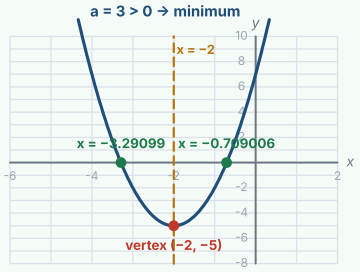
Example 2. **For that same function, is the y-intercept positive or negative?**

Solution.
y(0) = a(−3)(5) = −15a. With a < 0 this is **positive**. A negative a does not make the intercept negative — seven of your class assumed it did.

y(0) = a(−3)(5)	**= −15a**
a < 0	**so −15a > 0**
the intercept	**is positive**

Example 3. **y = 3x² + 12x + 7 has what kind of turning point?**

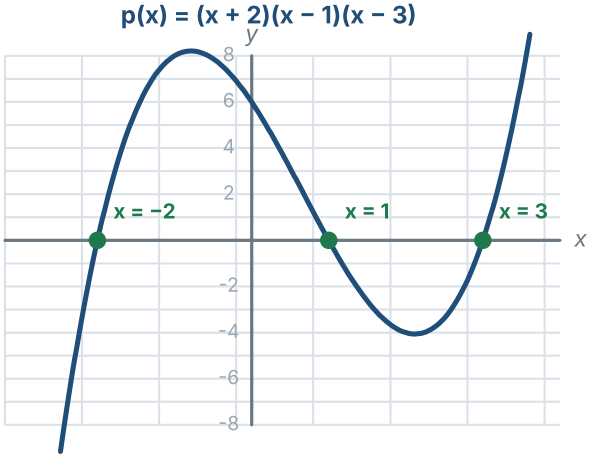
Solution.
a = 3 > 0, so a **minimum**.



4 Reading intercepts off a graph

x-intercepts are where y = 0 | the y-intercept is where x = 0

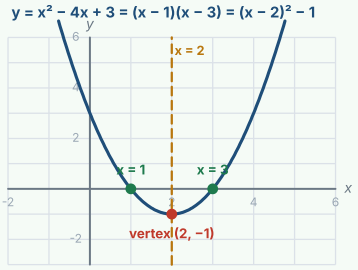
Substituting zero is the whole method. For the y-intercept of a product, multiply the constants: the y-intercept of (x + 2)(x − 1)(x − 3) is 2 × −1 × −3 = 6.



Where the curve meets the x-axis, y is zero; where it meets the y-axis, x is zero. Putting x = 0 into the three brackets gives the height 6.

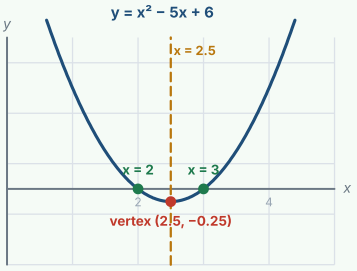
Example 1. **Find the y-intercept of y = (x − 1)(x − 3).**

Solution.
y(0) = (−1)(−3) = **3**.



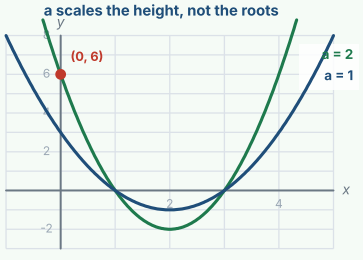
Example 2. **Find the x-intercepts of y = x² − 5x + 6.**

Solution.
(x − 2)(x − 3) = 0.
(2, 0) and (3, 0).



Example 3. **A parabola has x-intercepts 1 and 3 and passes through (0, 6). Find a in y = a(x−1)(x−3).**

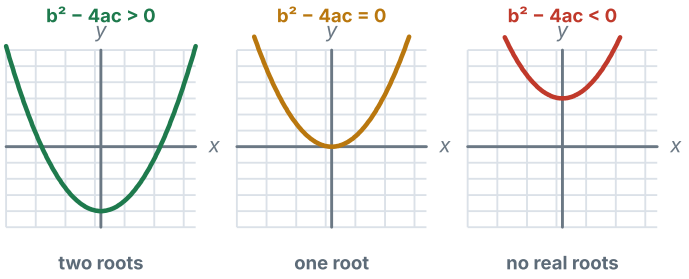
Solution.
6 = a(−1)(−3) = 3a, so a = **2**.



5 How many times does it cross?

the number of x-intercepts is decided by the discriminant $b^2 - 4ac$

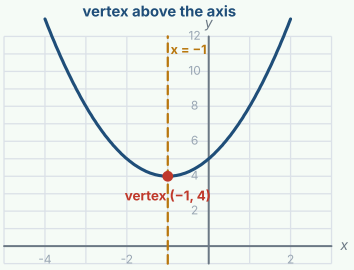
Two intercepts, one (a touch), or none. A question that says ‘touches the x-axis at exactly one point’ or ‘is tangent to the x-axis’ is asking you to set the discriminant to zero.



The same three cases as in 2.2, now read off the picture: cutting twice, grazing once, or missing the axis altogether.

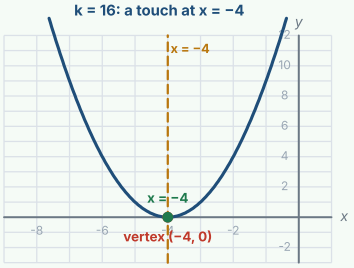
Example 1. **How many x-intercepts does $y = x^2 + 2x + 5$ have?**

Solution.
$4 - 20 = -16 < 0$.
None — the parabola sits entirely above the axis.



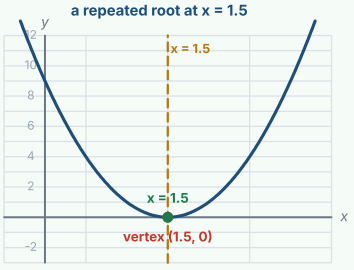
Example 2. **For what k does $y = x^2 + 8x + k$ touch the x-axis exactly once?**

Solution.
$64 - 4k = 0$.
$k = \mathbf{16}$.



Example 3. **How many x-intercepts does $y = 4x^2 - 12x + 9$ have?**

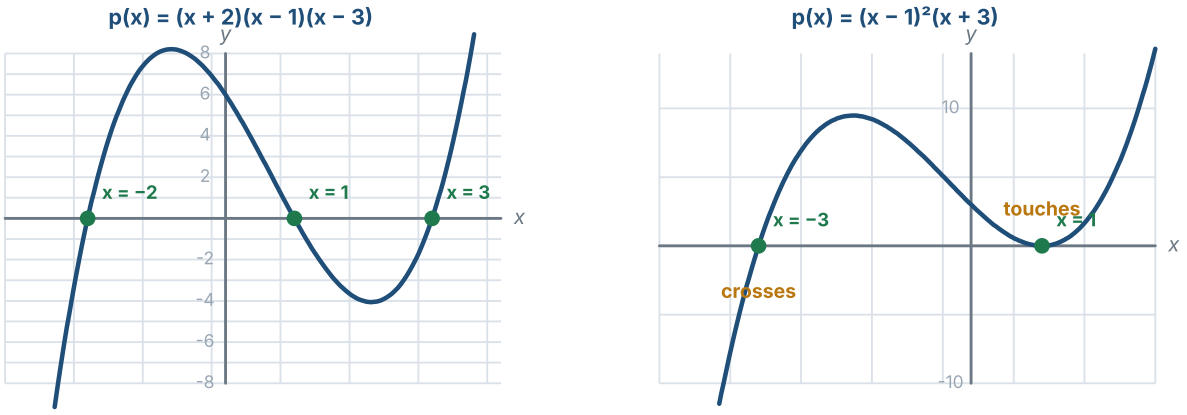
Solution.
$144 - 144 = 0$.
One, at the vertex $x = 1.5$.



6 Cubics and higher

the graph crosses at odd multiplicity and touches at even | degree n has at most n − 1 turning points

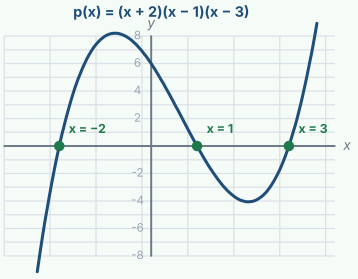
The factored form of a polynomial is a map of its graph. Each factor gives a zero, and the power on that factor says whether the curve passes through or bounces off.



Three single factors on the left: three crossings and two turning points. A squared factor on the right: the curve meets the axis at x = 1 and turns straight back.

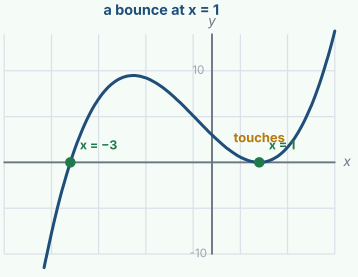
Example 1. **Describe the graph of y = (x + 2)(x − 1)(x − 3).**

Solution.
Three simple zeros, so it **crosses at −2, 1 and 3**, with at most 2 turning points.



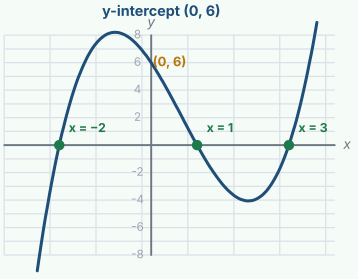
Example 2. **Describe the graph of y = (x − 1)²(x + 3) at x = 1.**

Solution.
Multiplicity 2, so the curve **touches the axis and turns back**.



Example 3. **What is the y-intercept of y = (x + 2)(x − 1)(x − 3)?**

Solution.
(2)(−1)(−3) = **6**.



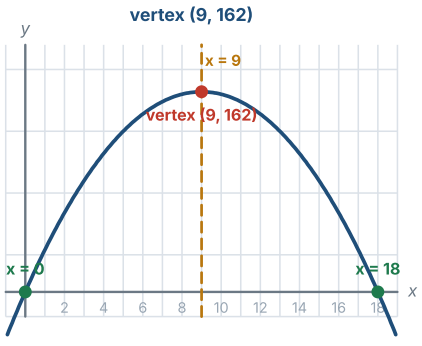
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

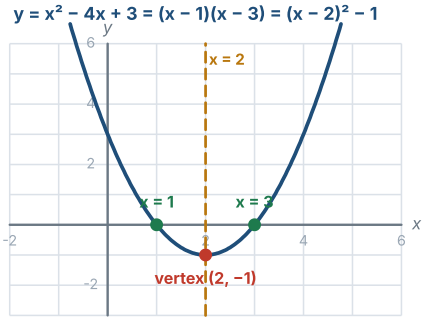
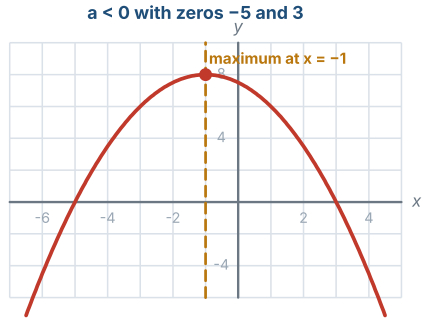
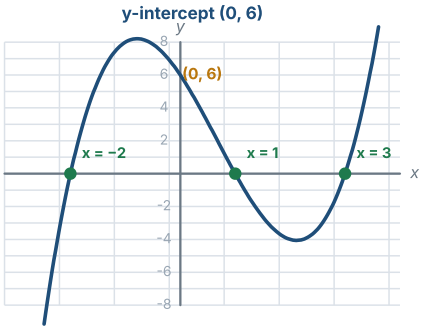
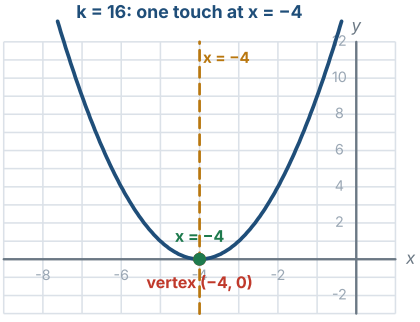
The SAT gives you Desmos on every maths question. This topic is **where Desmos earns its keep**. Graph the function, click the vertex and the intercepts, and the question usually answers itself.

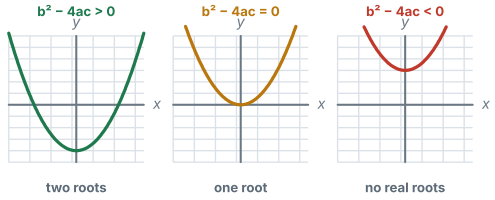
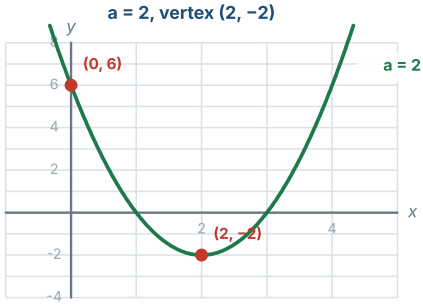
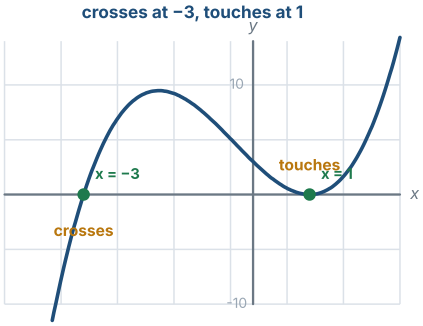
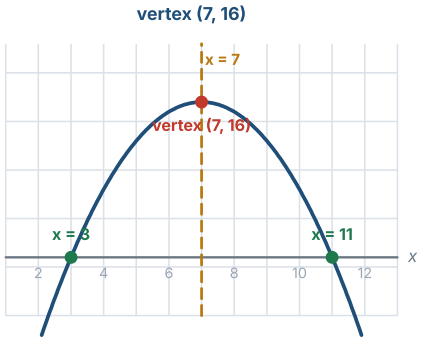
#	Question	Type into Desmos	Then	Answer
D1	Graph $y = x^2 - 4x + 3$. What is the vertex?	`y=x^2-4x+3`	Click the turning point; Desmos labels it.	(2, -1)
D2	For that graph, what are the x-intercepts?	`y=x^2-4x+3`	Click where the curve meets the axis.	1 and 3
D3	What is the minimum value of $y = x^2 - 6x + 5$?	`y=x^2-6x+5`	Read the y of the vertex.	-4
D4	$R(x) = -2x^2 + 36x$. What x gives the maximum?	`y=-2x^2+36x`	The x of the vertex, not the y.	9
D5	How many x-intercepts does $y = x^2 + 2x + 5$ have?	`y=x^2+2x+5`	The curve never reaches the axis.	0
D6	For what k does $x^2 + 8x + k = 0$ have exactly one solution?	`8^2/4`	Set the discriminant to zero.	16
D7	What is the y-intercept of $y = (x + 2)(x - 1)(x - 3)$?	`(0+2)*(0-1)*(0-3)`	Substitute x = 0.	6
D8	Graph $y = (x - 1)^2(x + 3)$. How many turning points does it have?	`y=(x-1)^2(x+3)`	Degree 3 allows at most 2.	2
D9	A parabola has x-intercepts 1 and 3 and passes through (0, 6). What is a in $y = a(x-1)(x-3)$?	`6/((0-1)*(0-3))`	Solve 3a = 6.	2
D10	What is the value of $-2x^2 + 36x$ at x = 9?	`-2*9^2+36*9`	The maximum revenue itself.	162

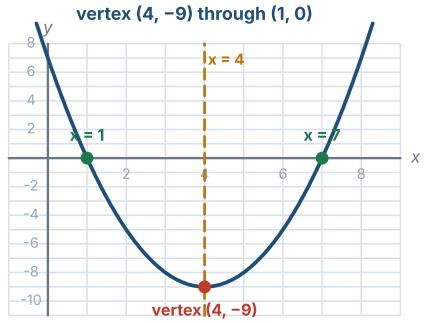
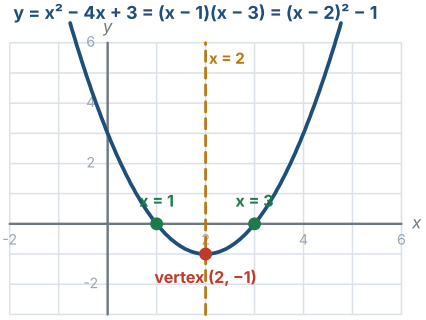
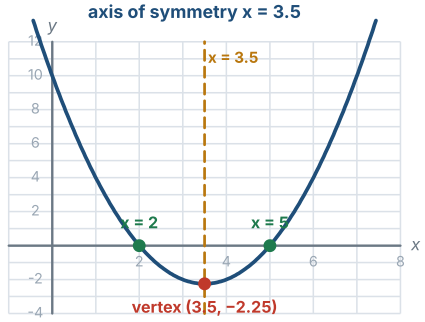
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Decide first whether the question wants **where** something happens (an x) or **how much** it is (a y). Most of the lost marks in this topic are the right number answering the wrong question.

#	Level	Question	Answer	Solution
P1	Hard	A candle shop's daily revenue, in dollars, is modelled by $R(x) = -2x^2 + 36x$, where x is the price of one candle in dollars. What price gives the maximum daily revenue? 	$9	The vertex sits at $x = -b/2a = -36/(-4) =$ **$9**. A common wrong answer is the maximum revenue R(9) = 162 — the answer to a different question. Five of your class chose C, reading 36 straight off the equation.

#	Level	Question	Answer	Solution
P2	Hard	The graph of $y = x^2 - 4x + 3$ has its vertex at which point? 	(2, −1)	$x = -b/2a = 4/2 = 2$, and $y = 4 - 8 + 3 = -1$. Vertex **(2, −1)**. A common wrong answer drops the sign on −b.
P3	Hard	A candle stall's weekly profit is modelled by $y = a(x - 3)(x + 5)$, where $a < 0$. Which must be true about the graph? 	The graph has a maximum at x = −1.	The zeros are 3 and −5, so the vertex is at their midpoint $x = -1$, and $a < 0$ makes it a **maximum**. The trap seven of your class took: the y-intercept is $a(-3)(5) = -15a$, which is *positive* when $a < 0$.
P4	Hard	What is the y-intercept of the graph of $y = (x + 2)(x - 1)(x - 3)$? 	6	Substitute $x = 0$: $(2)(-1)(-3) =$ **6**.
P5	Hard	For what value of k does the graph of $y = x^2 + 8x + k$ touch the x-axis at exactly one point? 	16	Touching once means the discriminant is zero: $64 - 4k = 0$. $k =$ **16**.

#	Level	Question	Answer	Solution
P6	Hard	How many x-intercepts does the graph of $y = x^2 + 2x + 5$ have? 	Zero	$b^2 - 4ac = 4 - 20 = -16 < 0$. The parabola opens upward with vertex above the axis, so it has **no** x-intercepts.
P7	Very hard	A parabola has x-intercepts at (1, 0) and (3, 0) and passes through (0, 6). What is the y-value of its vertex? 	−2	$y = a(x - 1)(x - 3)$, and $6 = a(-1)(-3) = 3a$ gives $a = 2$. The vertex sits at $x = 2$. $y = 2(1)(-1) = \mathbf{-2}$. A common wrong answer is the vertex y when $a = 1$.
P8	Very hard	The graph of a polynomial p crosses the x-axis at $x = -3$ and touches it at $x = 1$. Which could be p(x)? 	$(x + 3)(x - 1)^2$	Crossing needs odd multiplicity, touching needs even. $= \mathbf{(x + 3)(x - 1)^2}$. A common wrong answer swaps which zero is repeated; D has both signs wrong.
P9	Very hard	The function f is defined by $f(x) = -x^2 + 14x - 33$. What is the maximum value of f? 	16	The vertex is at $x = -14/(-2) = 7$. $f(7) = -49 + 98 - 33 = \mathbf{16}$. Answering 7 gives where the maximum occurs, not the maximum itself.

#	Level	Question	Answer	Solution
P10	Very hard	The graph of $y = a(x - h)^2 + k$ has its vertex at $(4, -9)$ and passes through $(1, 0)$. What is the value of a? 	1	$y = a(x - 4)^2 - 9$, and at $(1, 0)$: $0 = a(9) - 9$, so $9a = 9$. $a =$ **1**.
P11	Very hard	Two functions are given by $f(x) = x^2 - 4x + 3$ and $g(x) = (x - 2)^2 - 1$. State what is true here. 	f and g are the same function.	Expanding g: $x^2 - 4x + 4 - 1 = x^2 - 4x + 3 = f(x)$. They are **the same function written in two forms** — standard and vertex.
P12	Very hard	The graph of $y = x^2 + bx + 10$ has its axis of symmetry at $x = 3.5$. What is the value of b? 	−7	The axis of symmetry is $x = -b/2a = -b/2$. $-b/2 = 3.5$ gives $b =$ **−7**. A common wrong answer drops the minus sign.

The one habit to build. The x of the vertex is where the maximum occurs; the y is the maximum itself. On Test 5 five students answered 36 and one answered 162 when the question wanted 9 — all three numbers were in the problem, and only one answered what was asked.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify2.py; every figure is drawn to the numbers in the question.

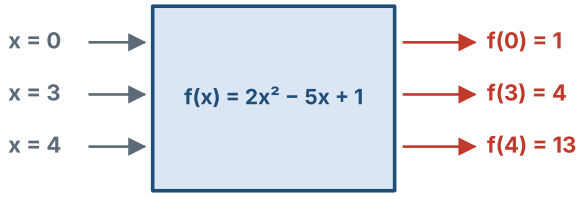
SAT Math 2.7 — Function Notation & Transformations

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Function notation

f(3) means put 3 in | f(x) = 3 means the output is 3 — solve for x

The brackets say substitute; they are not multiplication. f(a + b) is not f(a) + f(b), and the difference between f(3) = ? and f(x) = 3 is the difference between evaluating and solving.



put the number in, read the number out

two different jobs, one notation

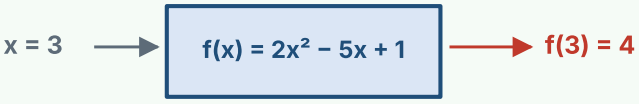
f(3) = ?	put 3 in, read the output
f(x) = 3	the output is 3, solve for x

The machine takes an input and returns an output. 'Find f(3)' runs it forwards; 'solve f(x) = 3' runs it backwards and can have two answers.

Example 1. **f(x) = 2x² − 5x + 1. Find f(3).**

Solution.

2(9) − 15 + 1 = 18 − 15 + 1 = **4**.



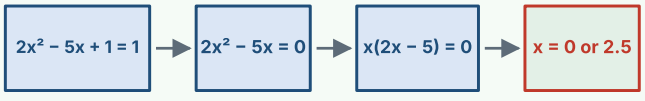
put the number in, read the number out

Example 2. **For that f, solve f(x) = 1.**

Solution.

2x² − 5x + 1 = 1 gives 2x² − 5x = 0, so x(2x − 5) = 0.

x = **0 or 2.5**.

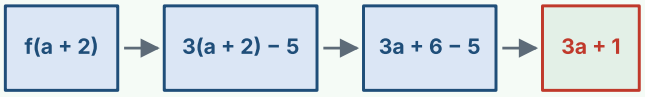


Example 3. **f(x) = 3x − 5. Find f(a + 2).**

Solution.

Substitute the whole expression: 3(a + 2) − 5.

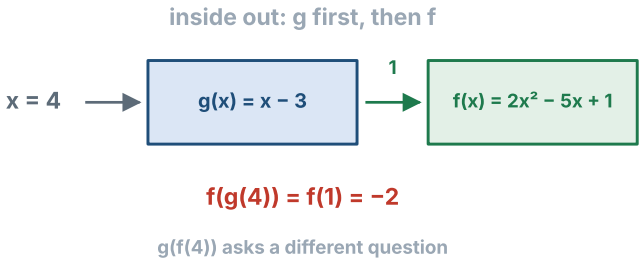
= **3a + 1**.



2 Composition

f(g(x)): work the inner function first, then feed that number outward

Composition is done from the inside out, and the order matters: f(g(2)) is usually not g(f(2)). Write the inner value down before touching the outer function.



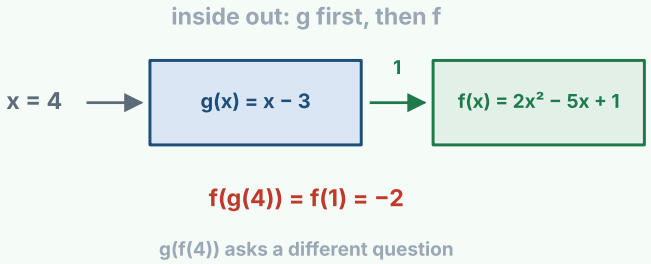
g(4) = 1 comes out of the first box and goes straight into the second, where f(1) = −2. Reversing the boxes gives a different number altogether.

Example 1. **f(x) = 2x² − 5x + 1 and g(x) = x − 3. Find f(g(4)).**

Solution.

g(4) = 1 first.

f(1) = 2 − 5 + 1 = **−2**.

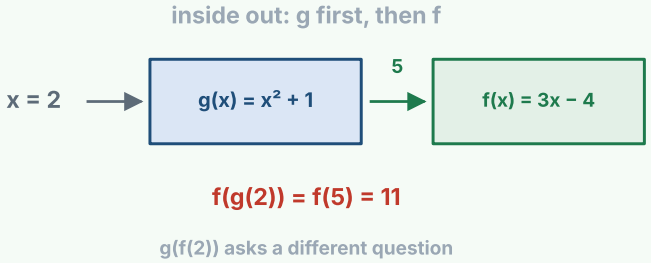


Example 2. **f(x) = 3x − 4 and g(x) = x² + 1. Find f(g(2)).**

Solution.

g(2) = 5.

f(5) = 15 − 4 = **11**.

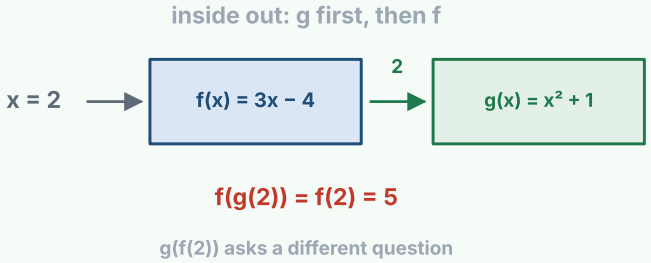


Example 3. **For those same functions, find g(f(2)).**

Solution.

f(2) = 2.

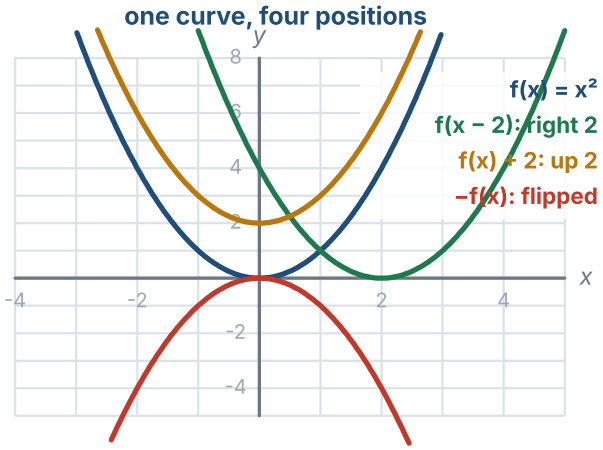
g(2) = 5. Different from f(g(2)) = 11 — **order matters**.



3 Vertical shifts and reflections

f(x) + k moves up k | f(x) − k moves down | −f(x) flips over the x-axis

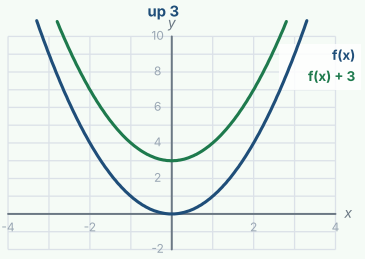
Changes made *outside* the function act on the output, so they behave exactly as you expect: add to go up, negate to flip vertically.



Everything outside the bracket does what it says: +2 lifts the curve, the minus sign turns it upside down. Only the green curve, whose change is inside the bracket, moves sideways.

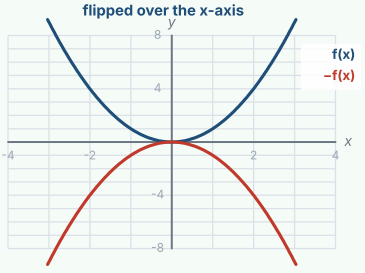
Example 1. **The graph of y = f(x) is shifted up 3. Write the new function.**

Solution.
y = f(x) + 3.



Example 2. **f(x) = x². Describe y = −f(x).**

Solution.
A reflection over the **x-axis**: the upward parabola becomes a downward one.



Example 3. **The point (2, 7) is on y = f(x). Where is it on y = f(x) − 4?**

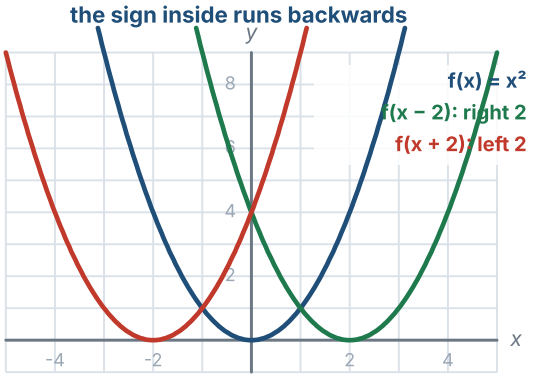
Solution.
Only the output changes.
(2, 3).

(2, 7) on f(x)	the input stays 2
f(x) − 4	the output drops by 4
new point	(2, 3)

4 Horizontal shifts — the opposite way

f(x − h) moves right h | f(x + h) moves left h

Changes made *inside* the brackets act on the input, and they run backwards. f(x − 2) shifts right, not left, because x must be 2 larger to produce the same output. This reversal is the single most-missed idea in transformations.



why inside runs backwards

f(x − 2)	x must be 2 bigger → right 2
f(x + 5)	x must be 5 smaller → left 5

To get the same output from f(x − 2) you must feed in an x that is 2 larger — so every point slides right. The minus sign moves the graph the opposite way to how it reads.

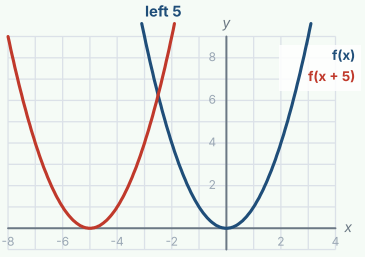
Example 1. **f(x) = x². Describe the graph of y = f(x − 2).**

Solution.
A shift **2 to the right**, giving y = (x − 2)² with vertex (2, 0).



Example 2. **Describe y = f(x + 5).**

Solution.
A shift **5 to the left** — the sign inside is the opposite of the direction.



Example 3. **The point (3, 8) is on y = f(x). Where is the matching point on y = f(x − 4)?**

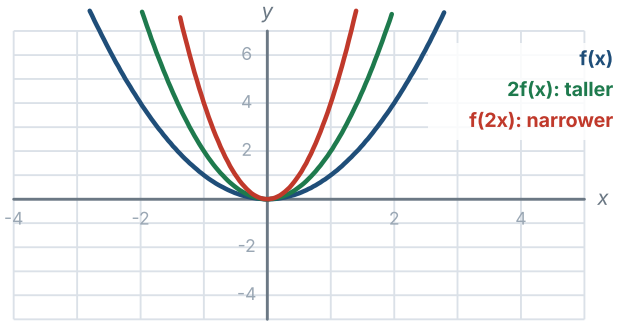
Solution.
The input must be 4 larger to give the same output.
(7, 8).

(3, 8) on f(x)	output 8 at input 3
f(x − 4)	needs x − 4 = 3
so x = 7	new point (7, 8)

5 Stretches

a·f(x) stretches vertically by a | f(bx) squeezes horizontally by 1/b

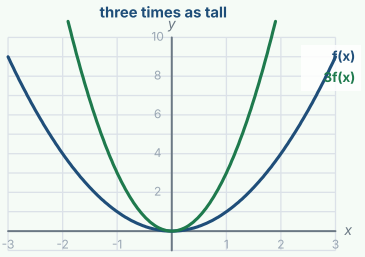
Outside multiplies the outputs, so heights change. Inside multiplies the inputs, so widths change — and again it runs backwards: f(2x) makes the graph half as wide, not twice.



Tripling outside makes every height three times as big. Doubling inside pulls the curve in to half its width, because x only has to reach half as far.

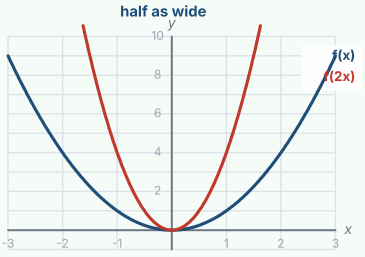
Example 1. **f(x) = x². Describe y = 3f(x).**

Solution.
Every y is tripled: a **vertical stretch by 3**, giving y = 3x².



Example 2. **Describe y = f(2x).**

Solution.
A **horizontal squeeze by a factor of 2** — the graph is half as wide.



Example 3. **The point (4, 6) is on y = f(x). Where is it on y = 2f(x)?**

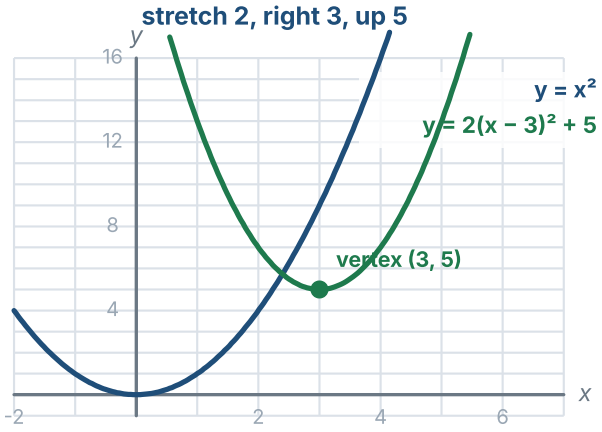
Solution.
The output doubles, the input does not.
(4, 12).

(4, 6) on f(x)	input 4, output 6
2f(x)	doubles the output only
new point	(4, 12)

6 Reading transformations from an equation

$y = a \cdot f(x - h) + k$ → stretch by a, right h, up k

Take the pieces in order and read the sign inside backwards. From vertex form $y = a(x - h)^2 + k$ you can state the whole transformation of $y = x^2$ in one sentence.



Read a, then h, then k: stretched by 2, moved 3 right (sign reversed) and 5 up. The vertex lands exactly at (h, k).

Example 1. **Describe $y = 2(x - 3)^2 + 5$ as a transformation of $y = x^2$.**

Solution.

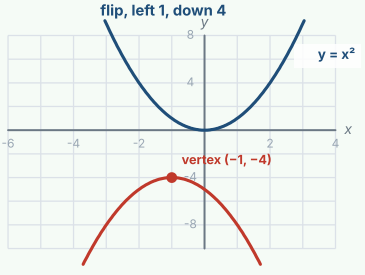
Stretched vertically by 2, moved 3 right and 5 up; vertex (3, 5).



Example 2. **Describe $y = -(x + 1)^2 - 4$.**

Solution.

Flipped over the x-axis, moved 1 left and 4 down; vertex (−1, −4).



Example 3. **$y = f(x)$ has a maximum at (2, 9). Where is the maximum of $y = f(x - 1) + 3$?**

Solution.

Right 1 and up 3.

(3, 12).

maximum (2, 9)	**on $y = f(x)$**
$f(x - 1)$	**moves it right 1 → $x = 3$**
$+ 3$	**moves it up 3 → $y = 12$**

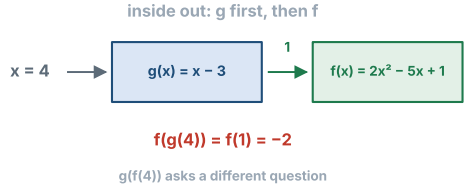
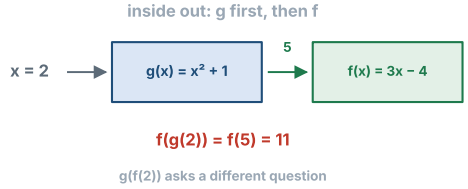
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

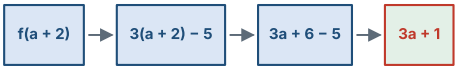
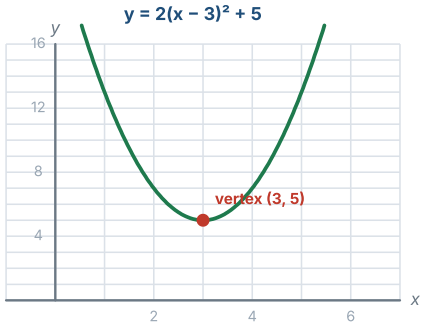
The SAT gives you Desmos on every maths question. Here it is **a transformation viewer**. Graph f and the transformed version together and watch which way the curve actually moves — it settles the inside-sign question for good.

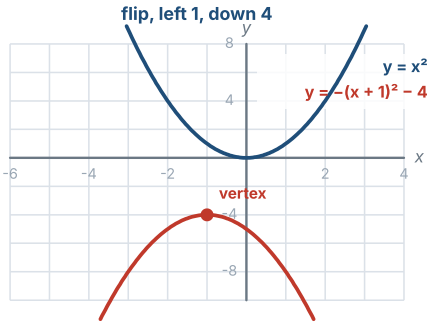
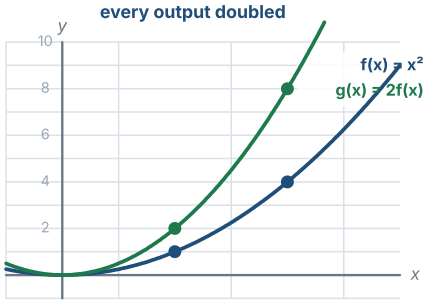
#	Question	Type into Desmos	Then	Answer
D1	$f(x) = 2x^2 - 5x + 1$. What is $f(3)$?	`2*3^2-5*3+1`	Substitute and evaluate in one line.	4
D2	$f(x) = 2x^2 - 5x + 1$ and $g(x) = x - 3$. What is $f(g(4))$?	`2*(4-3)^2-5*(4-3)+1`	Inner function first.	-2
D3	$f(x) = 3x - 4$ and $g(x) = x^2 + 1$. What is $f(g(2))$?	`3*(2^2+1)-4`	$g(2) = 5$, then $f(5)$.	11
D4	For those same functions, what is $g(f(2))$?	`(3*2-4)^2+1`	Different from D3 — order matters.	5
D5	Graph $y = x^2$ and $y = (x - 2)^2$. Which way did it move?	`y=(x-2)^2`	The sign inside runs backwards.	right 2
D6	What is the vertex of $y = 2(x - 3)^2 + 5$?	`y=2(x-3)^2+5`	Read h and k straight off.	(3, 5)
D7	$f(x) = 3x - 5$. What is $f(a + 2)$ when $a = 4$?	`3*(4+2)-5`	Substitute the whole expression.	13
D8	Solve $2x^2 - 5x + 1 = 1$.	`2x^2-5x+1=1`	Two solutions; the question asks where the output is 1.	0 and 2.5
D9	The point (2, 7) is on $y = f(x)$. What is the y-value on $y = f(x) - 4$?	`7-4`	Only the output shifts.	3
D10	What is the vertex of $y = -(x + 1)^2 - 4$?	`y=-(x+1)^2-4`	Inside sign flips; outside does not.	(-1, -4)

Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Sort each change into **outside** (acts on the output, behaves as expected) or **inside** (acts on the input, runs backwards) before you answer.

#	Level	Question	Answer	Solution
P1	Hard	The functions f and g are defined by $f(x) = 2x^2 - 5x + 1$ and $g(x) = x - 3$. What is the value of $f(g(4))$? 	−2	$g(4) = 1$ first, then $f(1) = 2 - 5 + 1 =$ **−2**. A common wrong answer is $f(4)$ — the inner function g was never applied.
P2	Hard	For $f(x) = 3x - 4$ and $g(x) = x^2 + 1$, what is the value of $f(g(2))$? 	11	$g(2) = 5$, then $f(5) = 15 - 4 =$ **11**. A common wrong answer works the functions in the wrong order: $g(f(2)) = 5$ is a different value again.

#	Level	Question	Answer	Solution
P3	Hard	The graph of y = f(x) is transformed to y = f(x − 2). Which describes the transformation? 	Shift 2 units right	A change inside the brackets acts on the input and runs *backwards*: x must be 2 larger to give the same output. The graph moves **2 right**. A common wrong answer is the instinctive one and the most common error in this topic.
P4	Hard	The point (2, 7) lies on the graph of y = f(x). Which point must lie on the graph of y = f(x) − 4? f(x) − 4 → outside → output drops 4 (2, 7) → becomes (2, 3) f(x − 4) → would give (6, 7) instead	(2, 3)	Subtracting outside the function changes only the output. **(2, 3).** A common wrong answer moves the input instead, which is what f(x − 4) would do.
P5	Hard	If f(x) = 3x − 5, which expression is equal to f(a + 2)? 	3a + 1	Substitute the whole expression: 3(a + 2) − 5 = 3a + 6 − 5. = **3a + 1**. A common wrong answer forgets to subtract the 5.
P6	Hard	What is the vertex of the graph of y = 2(x − 3)² + 5? 	(3, 5)	Vertex form y = a(x − h)² + k has vertex (h, k). **(3, 5)** — note h is +3 even though the bracket reads −3.
P7	Very hard	The graph of y = f(x) has a maximum at the point (2, 9). What are the coordinates of the maximum of y = f(x − 1) + 3? maximum (2, 9) → on y = f(x) f(x − 1) → right 1 → x = 3 + 3 → up 3 → y = 12	(3, 12)	Inside, −1 moves the graph right 1; outside, +3 moves it up 3. **(3, 12).** A common wrong answer moves left instead of right — the inside sign reverses.

#	Level	Question	Answer	Solution
P8	Very hard	Which describes the graph of $y = -(x + 1)^2 - 4$ as a transformation of $y = x^2$? 	Reflected over the x-axis, 1 left, 4 down	The minus outside reflects over the **x-axis**; (x + 1) moves **left** 1; −4 moves down 4. Vertex (−1, −4). A common wrong answer gets the direction of the shift backwards.
P9	Very hard	The point (4, 6) lies on the graph of $y = f(x)$. Which point must lie on the graph of $y = 2f(x)$? 2f(x) — outside → output doubles (4, 6) — becomes (4, 12) f(2x) — would halve the input instead	(4, 12)	Multiplying outside the function doubles the **output** only. **(4, 12).** A common wrong answer doubles the input, which is what f(x/2) would do.
P10	Very hard	If $f(x) = 2x^2 - 5x + 1$, what is the greatest value of x for which $f(x) = 1$? f(x) = 1 → 2x² − 5x = 0 → x(2x − 5) = 0 → x = 0 or 2.5	2.5	$2x^2 - 5x + 1 = 1$ gives $2x^2 - 5x = 0$, so $x(2x - 5) = 0$. x = 0 or 2.5, and the greater is **2.5**. Note this question asks you to *solve*, not to evaluate.
P11	Very hard	The graph of $y = f(x)$ passes through (0, 0), (1, 1) and (2, 4). The graph of $y = g(x)$ passes through (0, 0), (1, 2) and (2, 8). Which relationship is consistent with these points? 	g(x) = 2f(x)	Each output of g is exactly twice the matching output of f: 0, 2, 8 against 0, 1, 4. **g(x) = 2f(x)**, a vertical stretch. A common wrong answer would double the inputs, giving g(1) = f(2) = 4, not 2.
P12	Very hard	The function h is defined by $h(x) = f(x) + k$, where k is a constant. If $f(3) = 7$ and $h(3) = 2$, what is the value of k? h(3) = f(3) + k — 2 = 7 + k k = −5 — the graph moved down 5	−5	h(3) = f(3) + k, so 2 = 7 + k. k = **−5**.

The one habit to build. Inside the brackets, everything is the opposite of what it looks like: f(x − 2) goes right, f(x + 5) goes left, f(2x) gets narrower. Your class scored 13% on transformations, the lowest figure on the whole syllabus, and this single rule is most of it.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify2.py; every figure is drawn to the numbers in the question.

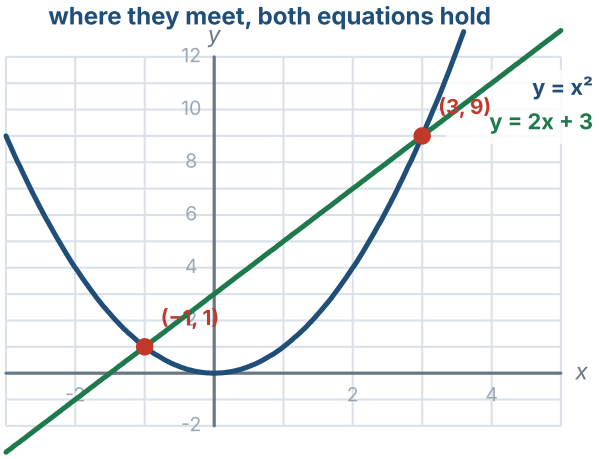
SAT Math 2.8 — Nonlinear Systems

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

1 Solving by substitution

set the two expressions for y equal | solve the resulting equation | find each y

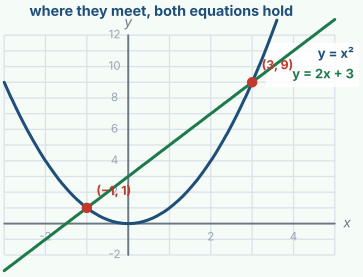
A nonlinear system is one curve and one line, or two curves. Where they meet, both equations hold, so equating them turns the pair into a single equation you already know how to solve. Do not stop at x — the solutions are points.



The two crossings are the two solutions. Equating $x^2 = 2x + 3$ finds their x-values; substituting back into either equation gives the matching y.

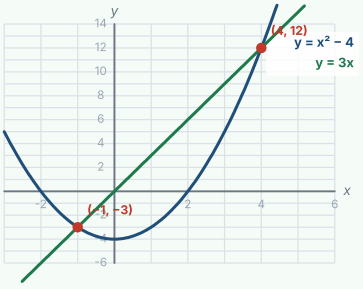
Example 1. **Solve $y = x^2$ and $y = 2x + 3$.**

Solution.
$x^2 = 2x + 3$, so $x^2 - 2x - 3 = 0$ and $(x - 3)(x + 1) = 0$.
$x = 3$ or -1, giving **(3, 9) and (−1, 1)**.



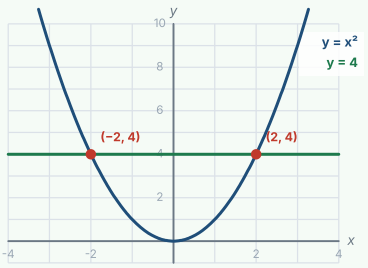
Example 2. **Solve $y = x^2 - 4$ and $y = 3x$.**

Solution.
$x^2 - 3x - 4 = 0$, so $(x - 4)(x + 1) = 0$.
(4, 12) and (−1, −3).



Example 3. **Solve $y = x^2$ and $y = 4$.**

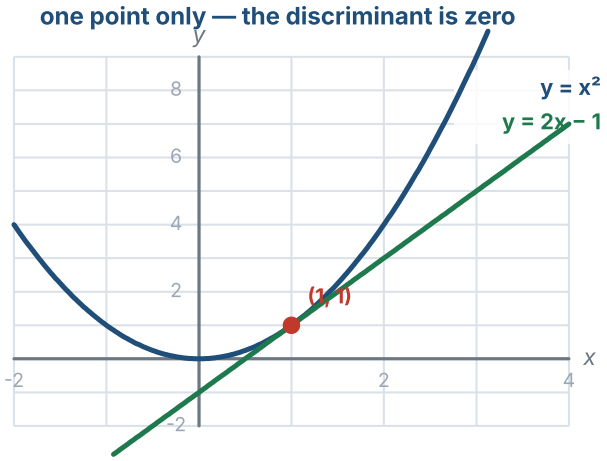
Solution.
$x^2 = 4$, so $x = \pm 2$.
(2, 4) and (−2, 4).



2 How many intersection points?

after substituting, the discriminant of the resulting quadratic counts the meetings

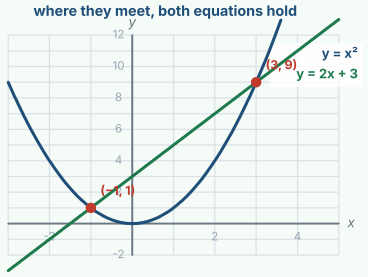
Two real solutions means the line cuts the curve twice; one means it is tangent; none means they miss entirely. This is the same discriminant test as always, wearing a different hat.



Substituting $y = 2x - 1$ into $y = x^2$ gives $x^2 - 2x + 1 = 0$ with discriminant 0 — and the picture shows the line just grazing the curve.

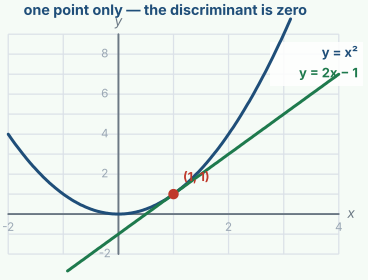
Example 1. **How many times does $y = 2x + 3$ meet $y = x^2$?**

Solution.
$x^2 - 2x - 3 = 0$ has discriminant $4 + 12 = 16 > 0$.
Twice.



Example 2. **How many times does $y = 2x - 1$ meet $y = x^2$?**

Solution.
$x^2 - 2x + 1 = 0$ has discriminant $4 - 4 = 0$.
Once — the line is tangent.

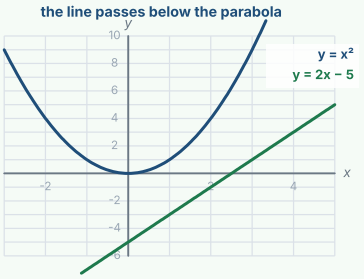


Example 3. **How many times does $y = 2x - 5$ meet $y = x^2$?**

Solution.

$x^2 - 2x + 5 = 0$ has discriminant $4 - 20 = -16 < 0$.

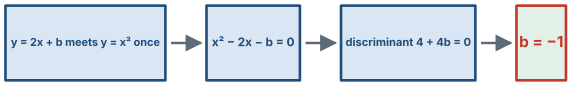
Never.



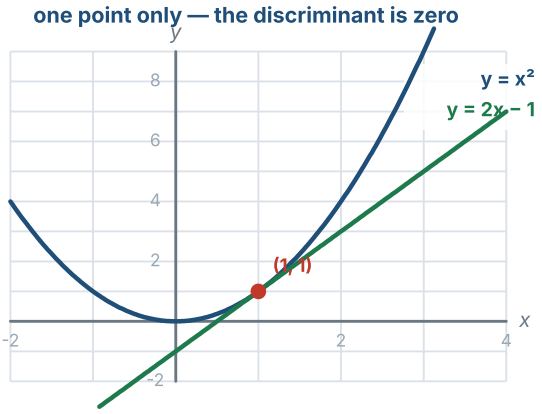
3 Tangency questions

‘meets at exactly one point’ ⇔ discriminant = 0

Whenever a question fixes a constant so that a line touches a curve once, substitute, collect into $ax^2 + bx + c = 0$, and set $b^2 - 4ac = 0$. Watch the sign at the last step — that is where the marks go.



the final sign is where five of your class lost the mark



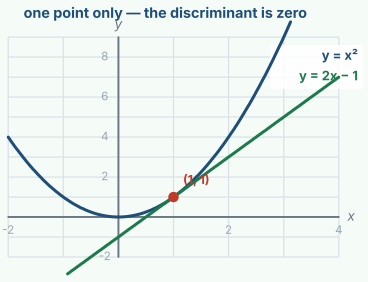
Set up, substitute, set the discriminant to zero, then solve for the constant — and check the sign of the answer against the picture.

Example 1. **For what b does $y = 2x + b$ meet $y = x^2$ exactly once?**

Solution.

$x^2 - 2x - b = 0$ with discriminant $4 + 4b = 0$.

$b = \mathbf{-1}$. Five of your class reached +1 here.

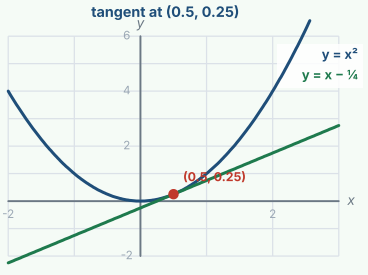


Example 2. **For what c does $y = x + c$ meet $y = x^2$ exactly once?**

Solution.

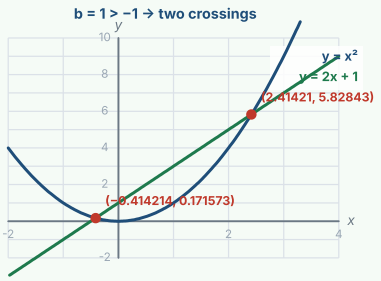
$x^2 - x - c = 0$, so $1 + 4c = 0$.

$c = \mathbf{-1/4}$.



Example 3. **Why does a positive b give two intersections in the first example?**

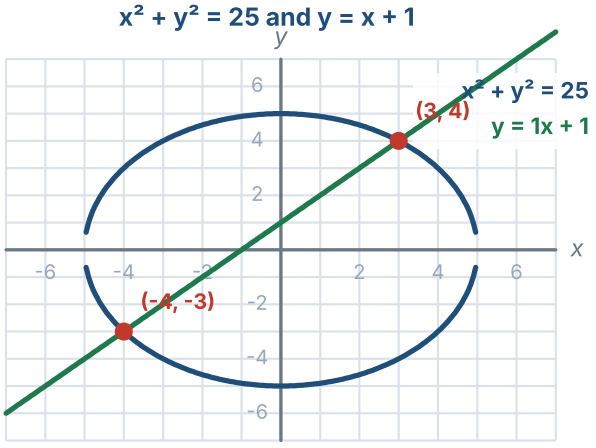
Solution.
$4 + 4b > 0$ whenever $b > -1$, so the discriminant is positive and the line cuts the parabola **twice**.



4 Circle and line systems

substitute the line into the circle equation, then solve the quadratic

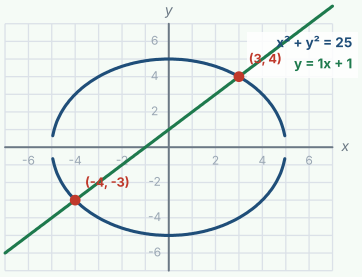
The method does not change when one equation is a circle. Replace y in $x^2 + y^2 = r^2$ using the line, expand carefully, and solve.



The line cuts the circle at two lattice points. Substituting $y = x + 1$ turns the system into $2x^2 + 2x - 24 = 0$, whose roots are those two x-values.

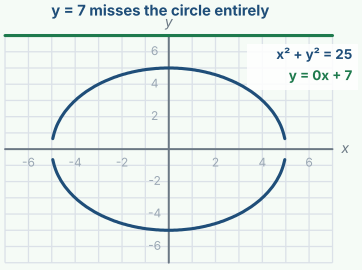
Example 1. **Solve $x^2 + y^2 = 25$ and $y = x + 1$.**

Solution.
$x^2 + (x + 1)^2 = 25$ gives $2x^2 + 2x - 24 = 0$, so $x^2 + x - 12 = 0$.
$x = 3$ or -4, giving **(3, 4) and (−4, −3)**.



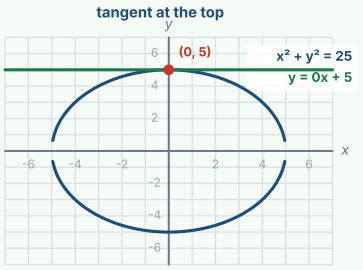
Example 2. **How many points does $y = 7$ share with $x^2 + y^2 = 25$?**

Solution.
$x^2 = 25 - 49 = -24$, which is impossible.
None — the line lies outside the circle.



Example 3. **How many points does y = 5 share with that circle?**

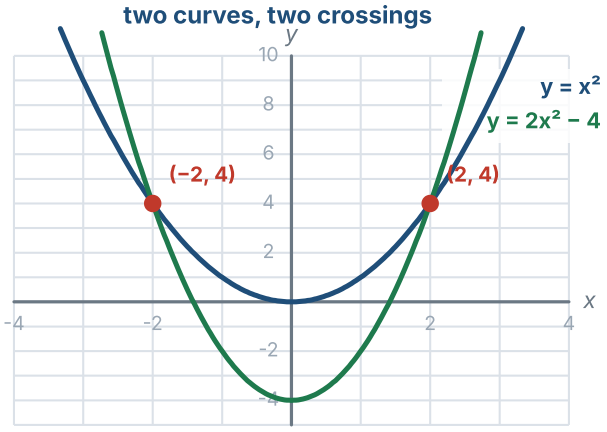
Solution.
$x^2 = 0$, so $x = 0$ only.
One: the line is tangent at (0, 5).



5 Systems with two curves

equate the two expressions | you may get a higher-degree equation

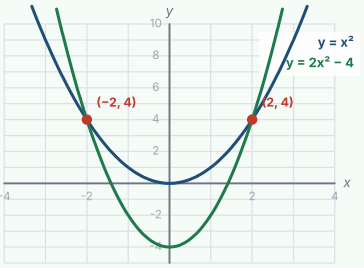
Two parabolas, or a parabola and a reciprocal, follow the same rule. The resulting equation may be cubic or quartic, so factor fully and keep every root.



Both curves are even, so the crossings come in a symmetric pair. Equating them gives $x^2 = 4$ and both roots count.

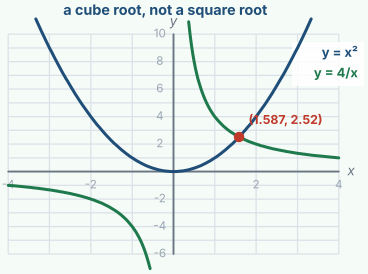
Example 1. **Solve $y = x^2$ and $y = 2x^2 - 4$.**

Solution.
$x^2 = 2x^2 - 4$ gives $x^2 = 4$.
(2, 4) and (−2, 4).



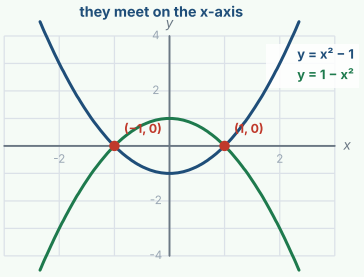
Example 2. **Solve $y = x^2$ and $y = 4/x$ for $x \neq 0$.**

Solution.
$x^2 = 4/x$, so $x^3 = 4$ and $x = \sqrt[3]{4} \approx 1.587$.
One real intersection, near **(1.587, 2.520)**.



Example 3. **Solve $y = x^2 - 1$ and $y = 1 - x^2$.**

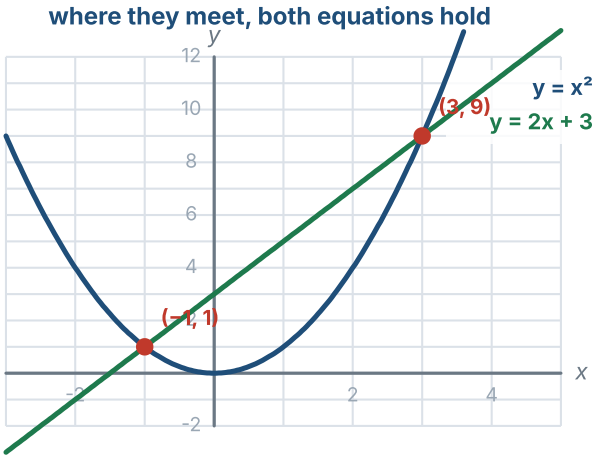
Solution.
$x^2 - 1 = 1 - x^2$ gives $2x^2 = 2$.
(1, 0) and (−1, 0).



6 Reading a system off a graph

a solution is a point the two graphs share | count intersections, not curves

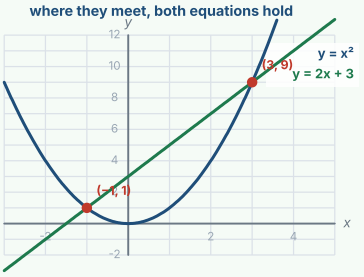
Desmos makes this almost free: enter both equations and click each crossing. The SAT often asks only for the number of solutions, or for one coordinate of one solution.



Two crossings means two solutions, and each one is a pair (x, y). Reading off only the x-values answers half the question.

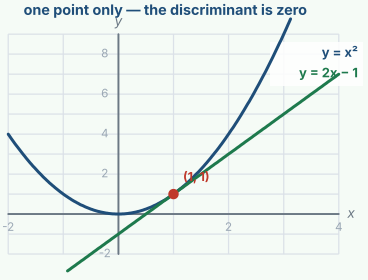
Example 1. **Two graphs cross at (3, 9) and (−1, 1). How many solutions does the system have?**

Solution.
Two — one for each shared point.



Example 2. **A line touches a parabola at one point. How many solutions does that system have?**

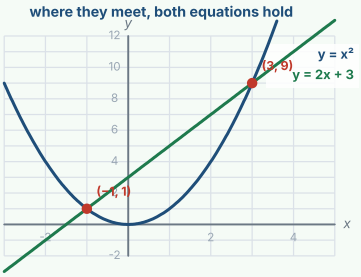
Solution.
One. A tangency is a single repeated solution.



Example 3. **A system's solutions are (3, 9) and (−1, 1). What is the sum of the x-coordinates?**

Solution.

3 + (−1) = **2**. This also equals −b/a for the quadratic you solved.



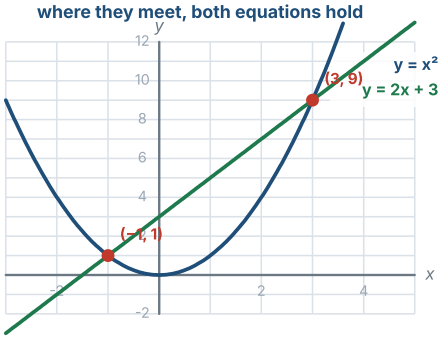
Part 1 — With Desmos (10 questions · draw the figure; the picture is faster than the algebra)

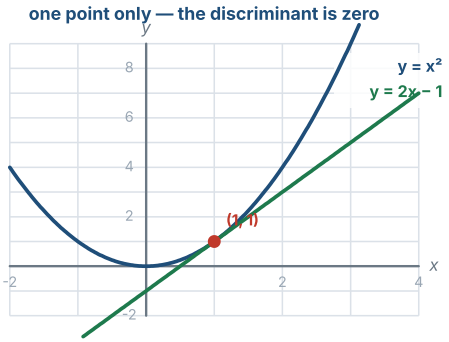
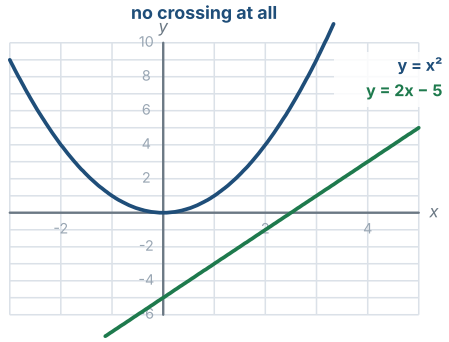
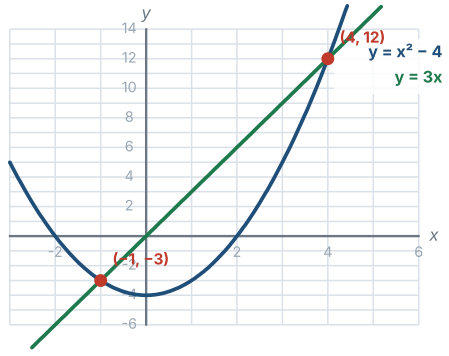
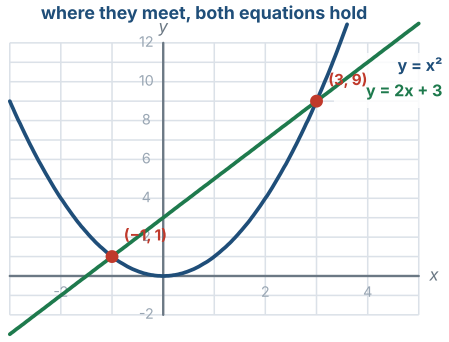
The SAT gives you Desmos on every maths question. Here it is **an intersection finder**. Enter both equations and click every crossing; Desmos gives you each point exactly.

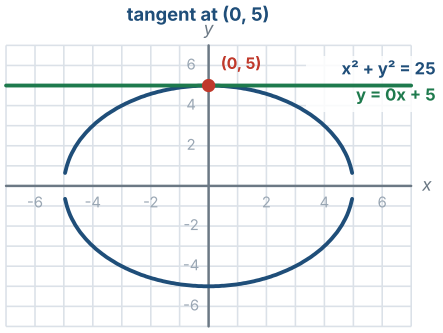
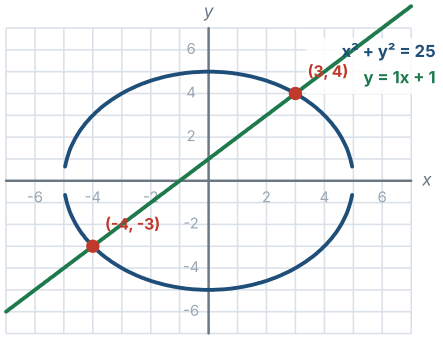
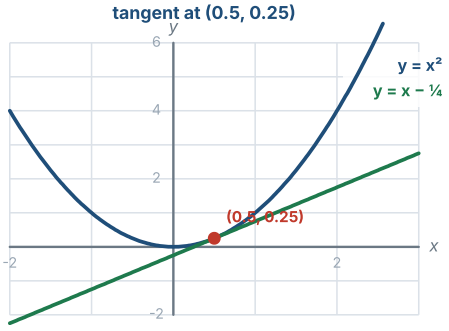
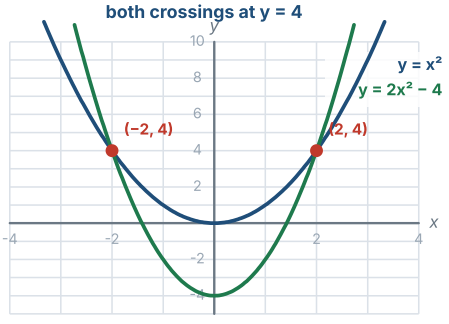
#	Question	Type into Desmos	Then	Answer
D1	Solve the system $y = x^2$ and $y = 2x + 3$. What is the larger x-value?	`y=x^2 and y=2x+3`	Graph both and click the intersections.	3
D2	For that system, what is the sum of the two x-values?	`3+(-1)`	Equals $-b/a$ for $x^2 - 2x - 3$.	2
D3	For what value of b does $y = 2x + b$ meet $y = x^2$ exactly once?	`-(2^2)/4`	Set $4 + 4b = 0$.	-1
D4	How many times does $y = 2x - 5$ meet $y = x^2$?	`y=x^2 and y=2x-5`	The line passes below the parabola.	0
D5	Solve $x^2 + y^2 = 25$ and $y = x + 1$. What is the larger x-value?	`x^2+(x+1)^2=25`	Substitute, then solve the quadratic.	3
D6	Solve $y = x^2$ and $y = 2x^2 - 4$. What are the x-values?	`x^2=2x^2-4`	Both curves are even, so the answers are symmetric.	2 and -2
D7	Solve $y = x^2$ and $y = 4/x$. What is x, to three decimal places?	`x^3=4`	A cube root, not a square root.	1.587
D8	How many points does the line $y = 5$ share with $x^2 + y^2 = 25$?	`25-5^2`	Zero under the root means tangency.	1
D9	Solve $y = x^2 - 4$ and $y = 3x$. What is the larger x-value?	`x^2-4=3x`	Factor $x^2 - 3x - 4$.	4
D10	For what value of c does $y = x + c$ meet $y = x^2$ exactly once?	`-1/4`	Set $1 + 4c = 0$.	-0.25

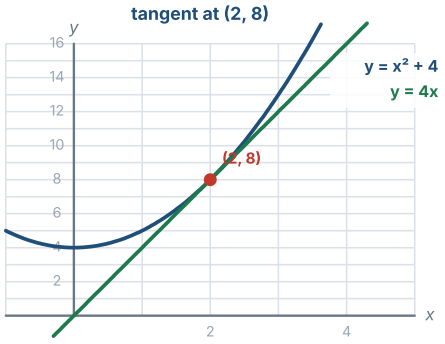
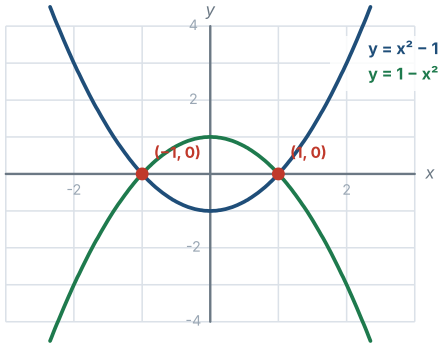
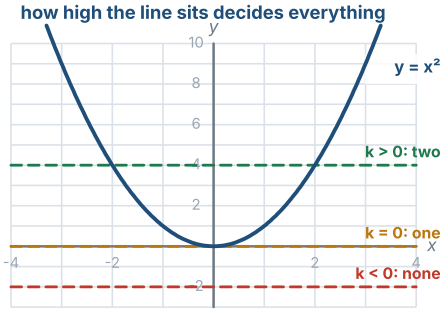
Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. When a question fixes a constant so two graphs meet **exactly once**, it is a discriminant question wearing a disguise.

#	Level	Question	Answer	Solution
P1	Hard	In the xy-plane, the graphs of $y = x^2$ and $y = 2x + 3$ intersect at two points. What is the sum of the x-coordinates of those points? 	2	$x^2 = 2x + 3$ gives $x^2 - 2x - 3 = 0$, whose roots sum to $-b/a$ = **2**. The roots are 3 and −1. A common wrong answer is the larger root alone.

#	Level	Question	Answer	Solution
P2	Hard	In the xy-plane, the line $y = 2x + b$, where b is a constant, intersects the parabola $y = x^2$ at exactly one point. What is the value of b? 	−1	$x^2 - 2x - b = 0$ must have discriminant zero: $4 + 4b = 0$. $b = \mathbf{-1}$. Five of your class answered +1 — the method was right and the sign was lost at the last step.
P3	Hard	How many solutions does the system $y = x^2$ and $y = 2x - 5$ have? 	Zero	$x^2 - 2x + 5 = 0$ has discriminant $4 - 20 = -16 < 0$. **No real solutions** — the line passes entirely below the parabola.
P4	Hard	The system $y = x^2 - 4$ and $y = 3x$ has two solutions. What is the greater of the two x-values? 	4	$x^2 - 3x - 4 = 0$ factors as $(x - 4)(x + 1) = 0$. $x = 4$ or -1, so the greater is **4**.
P5	Hard	Which ordered pair is a solution to the system $y = x^2$ and $y = 2x + 3$? 	(−1, 1)	Substituting $x = -1$: $y = 1$ from the parabola and $y = -2 + 3 = 1$ from the line, so both hold. **(−1, 1).** A common wrong answer uses the right y but the wrong sign of x.

#	Level	Question	Answer	Solution
P6	Hard	How many points of intersection do the line $y = 5$ and the circle $x^2 + y^2 = 25$ have? 	One	Substituting gives $x^2 + 25 = 25$, so $x^2 = 0$ and $x = 0$ only. **One** point, (0, 5): the line is tangent at the top of the circle.
P7	Very hard	The system $x^2 + y^2 = 25$ and $y = x + 1$ has two solutions. What is the greater of the two x-values? 	3	$x^2 + (x + 1)^2 = 25$ gives $2x^2 + 2x − 24 = 0$, so $x^2 + x − 12 = 0$ and $(x + 4)(x − 3) = 0$. $x = 3$ or $−4$, so the greater is **3**. A common wrong answer is the circle's radius, not a solution.
P8	Very hard	For what value of c does the line $y = x + c$ intersect the parabola $y = x^2$ at exactly one point? 	−1/4	$x^2 − x − c = 0$ must have discriminant zero: $1 + 4c = 0$. $c = \mathbf{-1/4}$. A common wrong answer drops the minus sign.
P9	Very hard	The graphs of $y = x^2$ and $y = 2x^2 − 4$ intersect at two points. What is the y-coordinate of both points? 	4	$x^2 = 2x^2 − 4$ gives $x^2 = 4$, so $x = ±2$. In either case $y = x^2 = \mathbf{4}$. Both points share the same height.

#	Level	Question	Answer	Solution
P10	Very hard	In the xy-plane, the line y = mx intersects the parabola y = x² + 4 at exactly one point, where m is a positive constant. What is the value of m? 	4	x² − mx + 4 = 0 needs m² − 16 = 0, so m = ±4. With m positive, m = **4**. A common wrong answer is m², stopped one step early.
P11	Very hard	The graphs of y = x² − 1 and y = 1 − x² intersect at how many points, and where? 	Two points, at (1, 0) and (−1, 0)	x² − 1 = 1 − x² gives 2x² = 2, so x = ±1, and then y = 1 − 1 = 0. **(1, 0) and (−1, 0)** — the two parabolas meet on the x-axis.
P12	Very hard	A system consists of y = x² and y = k, where k is a constant. For which values of k does the system have exactly two real solutions? 	k > 0	x² = k has two real solutions exactly when k is positive, one when k = 0, and none when k is negative. **k > 0.** The horizontal line must sit above the vertex.

The one habit to build. Set the discriminant to zero, then check the sign of your answer against the graph. On Test 5 ten students reached the right equation for b and five of them wrote +1 instead of −1.

All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify2.py; every figure is drawn to the numbers in the question.